Gradient Subspace-Regularized Hyperspectral Image and Stripe-Coupled Nonconvex Tensor Low-Rank Priors for Destriping and Denoising

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***Abstract*— In this article, we propose a novel, unified, and effective hyperspectral image (HSI) destriping and denoising method with gradient subspace-regularized HSI and stripe-coupled nonconvex tensor low-rank priors (GSHSNTLRs). First, by exploiting the mode-3 low-rank properties of the gradients of HSI (i.e., the spatial horizontal gradient, spatial vertical gradient, and spectral gradient of HSI) along the spectral dimension, we apply the mode-3 low-rank decomposition of the gradients of HSI to obtain their representation coefficient tensors (RCTs), and further study the tensor low-tubal-rank properties of the RCTs in the gradient subspace. Thus, we propose the unified gradient subspace-regularized log tensor nuclear norm (LogTNN)-based nonconvex tensor low-rank prior term of the RCTs. Moreover, by fully considering the structural speciality of stripe noise, which has strong tensor low-tubal-rank property, we particularly study the HSI-guided tensor low-rank modeling for the stripe noise by exploring the tensor low-tubal-rank property of HSI plus stripe and propose the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term of HSI and stripe simultaneously. Subsequently, the proposed GSHSNTLR model is solved by using the alternating direction method of multipliers (ADMMs). Finally, lots of experimental results and analysis fully demonstrate the destriping and denoising performance and superiority of GSHSNTLR.**

***Index Terms*— Gradient subspace, hyperspectral image (HSI) destriping and denoising, low-rank decomposition, nonconvex tensor low-rank prior, representation coefficient tensors (RCTs), spatial and spectral gradients.**

NOMENCLATURE

Notations	*Definitions*
a, $\mathbf{a}$, $\mathbf{A}$, and $\mathcal{A}$	Scalar, vector, matrix, and tensor, respectively.
$\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Clean HSI (an order-3 tensor).
$\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Noisy HSI (an order-3 tensor).
$\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Stripe noise (an order-3 tensor).
$\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$	Sparse noise (an order-3 tensor) (i.e., impulse noise and deadlines).
$\mathcal{X}(i_1, i_2, i_3)$	(i_1, i_2, i_3)th element of $\mathcal{X}$.
$\mathcal{X}(:, :, i_3)$	i_3th frontal slice of $\mathcal{X}$.
$\text{unfold}_3(\mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$	Mode-3 unfolding of $\mathcal{X}$.
$\mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$	Mode-3 unfolding of $\mathcal{X}$.
$\lvert \cdot \rvert$	Absolute value operator.
$\text{rank}(\cdot)$	Rank operator of matrix.
$\lVert \cdot \rVert_*$	Nuclear norm operator of matrix.
$\lVert \mathcal{X} \rVert_1 = \sum_{i_1,i_2,i_3} \lvert \mathcal{X}(i_1, i_2, i_3) \rvert$	l_1-norm of $\mathcal{X}$.
$\lVert \mathcal{X} \rVert_F = \sqrt{\sum_{i_1,i_2,i_3} \lvert \mathcal{X}(i_1, i_2, i_3) \rvert^2}$	Frobenius norm of $\mathcal{X}$.
$(\cdot)^{\text{T}}$	Transpose operator.

I. INTRODUCTION

DUE to the inherent, abundant, and important spectral information of the hyperspectral image (HSI), the high-quality HSI is urgently demanded and also widely applied for such applications as object detection, environmental monitoring, classification, military surveillance, ecological protection [1], etc. However, the really collected HSI is easily and inevitably contaminated by a mixture of various types of noise, such as Gaussian noise, impulse noise, deadlines, and stripe noise, which not only seriously degrades the spatial–spectral image quality of HSI, but also seriously limits the subsequently high-level applications of HSI. Thus, to effectively remove the hybrid noise from the noisy HSI, it is crucial and also challenging to develop effective HSI denoising approaches to enhance the spatial–spectral image quality of HSI for the subsequently high-level applications.

Over past decades, various kinds of approaches have been developed for HSI denoising, such as transform-based and filter-based approaches, prior model-based approaches, and deep learning-based approaches [2].

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A. Related Work: Deep Learning-Based Approaches

Based on the rapid developments as well as the strong nonlinear mapping learning capabilities of deep learning, the deep learning-based approaches are widely used for HSI denoising recently. Among them, there are many representative approaches by using various different network architectures, such as convolutional neural network-based, U-Net-based, attention-based, transformer-based, and deep priors-based approaches [3], [4], [5], [6], [7], [8], [9], [10], [11]. Specifically, these deep learning-based approaches can generally perform promising and advanced denoising results for HSI denoising. Nevertheless, their generalization capacities are relatively weaker, which can be further interpreted and improved by fully considering the complex degradation process of HSI as well as different attributes of mixed noise in real world.

B. Related Work: Prior Model-Based Approaches

In this article, we mainly focus on investigating the prior model-based HSI denoising approaches, whose core idea is to formulate HSI denoising as an optimization model, i.e.,

$$\hat{\mathcal{X}} = \arg\min_{\mathcal{X}} \mathcal{L}(\mathcal{Y}, \mathcal{X}) + \alpha \mathcal{R}(\mathcal{X}) \tag{1}$$

where $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the noisy HSI containing the Gaussian noise, impulse noise, deadlines, and stripe noise, $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ denotes the clean HSI to be restored, $\mathcal{L}(\mathcal{Y}, \mathcal{X})$ denotes the loss term between $\mathcal{Y}$ and $\mathcal{X}$, $\mathcal{R}(\mathcal{X})$ denotes the prior term of $\mathcal{X}$, and α denotes the regularization parameter.

To the best of authors' knowledge, the recently mainstream prior modeling schemes of HSI mainly aim to investigate the latent spatial and spectral structural properties of HSI, such as local smoothness priors, low-rankness priors, and their combinations.

1) Local Smoothness Priors: For the local smoothness priors-based HSI denoising approaches, they mainly exploit the spatial and spectral smoothnesses between the adjacent pixels of HSI in the gradient domain by the total variation (TV) regularizers. Specifically, the representative ones are spectral–spatial adaptive hyperspectral TV (SSAHTV) [12], spatiospectral TV (SSTV) [13], enhanced 3-D TV (E-3DTV) [14], representative coefficient TV (RCTV) [15], and panchromatic-weighted RCTV (PWRCTV) [16].

2) Low-Rankness Priors: For the low-rankness priors-based HSI denoising approaches, they aim to investigate and exploit the inherently spectral and spatial low-rank structural properties of HSI as the matrix-based low-rankness priors or tensor-based low-rankness priors, which are summarized as follows.

1) *Matrix-Based Low-Rankness Priors:* Indeed, the HSI data have not only strong spectral correlations, but also the obvious 2-D patches-based spatial nonlocal similarity and the 3-D patches-based spatial–spectral nonlocal similarity. Thus, the matrix-based low-rankness priors aim to either unfold the 3-D HSI data along the spectral dimension into the 2-D low-rank matrix or reshape the similar groups of 2-D and 3-D nonlocal patches into the 2-D low-rank matrices for low-rankness modeling. Obviously, the representative ones are mainly based on the low-rank matrix decomposition, matrix-based nuclear norms, or nonconvex low-rank regularizers [17], [18], [19], [20], [21], [22].
2) *Tensor-Based Low-Rankness Priors:* More actually, the HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the 3-D data (i.e., the order-3 tensor), which has strong spatial and spectral structural correlations inherently. However, these aforementioned matrix-based low-rankness priors employing low-rank matrix decomposition, matrix-based nuclear norms, or nonconvex low-rank regularizers need to either unfold the 3-D HSI data along the spectral dimension into the 2-D low-rank matrix or reshape the similar groups of 2-D and 3-D nonlocal patches into the 2-D low-rank matrices for low-rankness modeling, which inevitably and seriously destroy the spatial and spectral structural correlations of the 3-D HSI data. Thus, more recent research works mainly focus on treating the HSI as the order-3 tensor to preserve the inherent 3-D structures and spatial and spectral structural correlations of HSI, and particularly apply the tensor representation modeling theory to develop many tensor-based low-rankness priors, which currently become the mainstream priors-based methods for HSI denoising. Obviously, the representative ones are mainly based on the low-rank tensor decomposition, tensor nuclear norms (TNNs), or nonconvex tensor low-rank regularizers [23], [24], [25], [26], [27], [28], [29], [30], [31], [32]. More specifically, among them, the methods [23], [24] exploit the nonlocal self-similarity of 3-D patches of HSI to form the similar patch group-based tensors, and then, respectively, perform the nonlocal low-rank regularized CANDECOMP/PARAFAC (CP) tensor decomposition [23] and the nonlocal tensor-ring approximation [24] on these similar patch group-based tensors. Moreover, the methods both exploit the low-rank Tucker decomposition of HSI directly [25], [26], and the method exploits the low-rank tensor-ring decomposition of HSI in the transformed domain [27]. Besides, the method exploits the TNN prior of HSI directly [28], and the methods both perform the nonlocal low-rank approximation of the similar patch group-based tensors of HSI by using the TNN [29] or weighted TNN [30]. Zheng et al. [31] exploit the low-tensor fibered rank prior of HSI and propose the three-directional TNN (3DTNN). Lin et al. [32] propose the nonconvex low-tensor-Tucker-rank prior term of HSI. However, these methods only consider the tensor-based low-rankness priors of HSI so that their performance can be further improved.

3) Combinations of the Low-Rankness and Local Smoothness Priors: Considering that either the local smoothness priors or the low-rankness priors cannot comprehensively model the local smoothness and low-rankness properties of HSI data at the same time, the combinations of low-rankness and local smoothness priors-based HSI denoising approaches have been also developed recently. Specifically, they not only exploit the inherently spectral and spatial low-rank structures

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of HSI, but also simultaneously use the spectral and spatial local smoothness of HSI, which also become more mainstream and comprehensive priors-based methods for HSI denoising recently. Obviously, the representative ones are the composite TV-regularized low-rank regularizers [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], and the joint low rankness and local smoothness-based regularizers, which enforce joint low rankness and local smoothness in a unified prior term [43], [44], [45].

C. Motivations

Although the recently mainstream HSI denoising methods with tensor-based low-rankness priors have achieved considerable success, there are still many difficulties and challenges that need to be further addressed, which are mainly summarized as follows.

1) For the tensor-based low-rankness priors of HSI, most of them are usually convex, where the tensor-based nonconvex low-rankness priors are inadequately exploited recently. More clearly, the nonconvex low-rankness priors can provide better results than the convex low-rankness priors. Thus, it is more urgent to study the tensor-based nonconvex low-rankness priors of HSI.
2) Moreover, some HSI denoising methods exploit the gradient-related characteristics of HSI for modeling, such as, E-3DTV [14], RCTV [15], and 3-D correlated TV (3DCTV) [43], where E-3DTV imposes the sparsity on the subspace bases of the unfolded matrices of spatial and spectral gradients along the spectral mode of HSI [14], RCTV imposes the 2-D spatial TV-based local smoothness prior of the representative coefficient obtained by the spectral low-rank decomposition of HSI [15], and 3DCTV imposes the nuclear norm-based low-rankness prior on the unfolded matrices of spatial and spectral gradients along the spectral mode of HSI [43]. However, the spectral low-rankness as well as the spectral subspace decomposition of the spatial and spectral gradients of HSI are not exploited simultaneously, and their representative coefficients are also tensors whose tensor-based low-rankness characteristics are not fully exploited either, which need to be further and deeply exploited.
3) Specifically, these works mainly focus on exploring and modeling the structural properties of HSI as well as the spatial and spectral correlations of HSI, and typically model the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ as also the sparse noise $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ by using the l_1-norm-based sparse prior (i.e., $\|\mathcal{S}\|_1$), while ignoring the modeling of structural and directional properties of stripe noise. In fact, the stripe noise has strong structural and directional properties [46], [47], [48], [49], [50], [51], [52], [53], [54], which is quite different from other types of sparse noise, such as impulse noise and deadlines. Thus, it is more reasonable to study the independent and contrapuntal structural prior modeling of the stripe noise separated from the sparse noise.
4) Although some recent works begin to model the stripe noise $\mathcal{B}$ separated from the spare noise $\mathcal{S}$ for simultaneous HSI destriping and denoising, they mainly model the stripe noise band-by-band by applying the matrix-based low-rank priors [49], [50], [51], [52], i.e., $\sum_{i_3=1}^{n_3} \text{rank}(\mathcal{B}(:,:,i_3))$ or $\sum_{i_3=1}^{n_3} \|\mathcal{B}(:,:,i_3)\|_*$. More clearly, the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is surely an order-3 tensor as the HSI tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. However, the stripe noise is simply modeled by the band-by-band matrix low-rank priors, which seriously ignore the tensor-based low-rankness prior modeling of the stripe noise $\mathcal{B}$. Therefore, it is more urgent to study the tensor-based low-rankness prior modeling of the stripe noise $\mathcal{B}$ as the HSI tensor $\mathcal{X}$ under the unified tensor modeling framework.
5) Although several HSI destriping and denoising methods also model the tensor-based low-rankness prior modeling of the stripe noise $\mathcal{B}$ and the HSI $\mathcal{X}$ separately under the unified tensor modeling framework, they always model the tensor low-rankness prior of the HSI as a tensor-based low-rankness prior term and model the tensor low-rankness prior of the stripe noise as another tensor-based low-rankness prior term [53], [54], where the former method models a unified nonconvex tensor low-rank prior term of HSI and HSI spectral gradient, and another unified nonconvex tensor-low rank prior term of stripe and the stripe vertical gradient [53], and the latter method models a tensor low-rank prior term of HSI and another tensor low-rank prior term of stripe separately by using the low-rank Tucker decompositions of the HSI and stripe separately [54]. However, the joint tensor low-rankness prior modeling of the HSI and the stripe noise in a unified tensor-based low-rankness prior term (not in previously two or more separate tensor-based low-rankness prior terms of the HSI and the stripe noise) is not investigated recently, which is desirable to be deeply investigated.

D. Our Contributions

In this article, to address these deficiencies for simultaneous HSI destriping and denoising, we particularly treat the stripe noise separated from the sparse noise for independent and contrapuntal structural modeling, and use the unified tensor representation modeling for the HSI, sparse noise, stripe noise, and Gaussian noise. Thus, from the perspectives of the unified tensor representation modeling and nonconvex tensor low-rankness modeling of HSI and stripe noise, we propose a novel, unified, and effective HSI destriping and denoising method with gradient subspace-regularized HSI and stripe-coupled nonconvex tensor low-rank priors (GSHSNTLRs). Specifically, the main contributions of this article can be summarized as follows.

1) First, by exploiting the mode-3 low-rank properties of the gradients of HSI (i.e., the spatial horizontal gradient, spatial vertical gradient, and spectral gradient of HSI) along the spectral dimension, we apply the mode-3 low-rank decomposition of the gradients of HSI to obtain their representation coefficient tensors (RCTs), and further study the tensor low-tubal-rank properties

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of the RCTs in the gradient subspace. Thus, we propose the unified gradient subspace-regularized log TNN (LogTNN)-based nonconvex tensor low-rank prior term of the RCTs.

2) Moreover, by fully considering the structural specialities of stripe noise, which is modeled as a third-order tensor having strong tensor low-tubal-rank property, we particularly study the HSI-guided tensor low-rank modeling for the stripe noise by exploring the tensor low-tubal-rank property of HSI plus stripe, and propose the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term of HSI and stripe simultaneously.
3) Based on the unified gradient subspace-regularized LogTNN-based nonconvex tensor low-rank prior term of the RCTs as well as the HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term of HSI and stripe, we propose a novel, unified, and effective GSHSNTLR model for HSI destriping and denoising. Subsequently, the proposed GSHSNTLR model is solved by using the alternating direction method of multipliers (ADMMs). Finally, lots of experimental results and analysis fully demonstrate the destriping and denoising performance and superiority of GSHSNTLR.

The rest parts of this article are arranged as follows. We propose the GSHSNTLR model in Section II and solve the proposed GSHSNTLR model in Section III. Section IV displays the overall experiments and analysis in detail. At last, the conclusion is shown in Section V.

II. Proposed GSHSNTLR Model

A. Problem Formulation

At first, Nomenclature gives many key notations for analysis.

Specifically, the problem of HSI destriping and denoising is to restore the clean HSI from its noisy HSI polluted by the sparse noise (i.e., impulse noise and dead lines), stripe noise, and Gaussian noise.

More different from most of the previous works, which treat the stripe noise as also the sparse noise for modeling, in this article, we particularly treat the stripe noise separated from the sparse noise for independent and contrapuntal modeling, and use the unified tensor representation modeling for the HSI, sparse noise, stripe noise, and Gaussian noise.

Therefore, the noisy HSI polluted by the sparse noise (i.e., impulse noise and deadlines), stripe noise, and Gaussian noise can be mathematically modeled as

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{B} + \mathcal{G} \tag{2}$$

where $\mathcal{Y} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the noisy HSI tensor, $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is the clean HSI tensor, $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, and $\mathcal{G} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ are, respectively, the additive sparse noise tensor, stripe noise tensor, and Gaussian noise tensor.

B. Gradient Subspace-Regularized RCT Nonconvex Low-Rank Prior of HSI

For the HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, it is clear that HSI $\mathcal{X}$ has strong spectral correlations modeled by the low-rank property in the spectral mode, namely, the mode-3 unfolding of $\mathcal{X}$, i.e., $\text{unfold}_3(\mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$ (or $\mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$), is low rank, where $\text{rank}(\mathbf{X}_{(3)}) = r \ll n_3$ denotes the rank of $\mathbf{X}_{(3)}$, and $\text{rank}(\cdot)$ is the rank operator of the matrix.

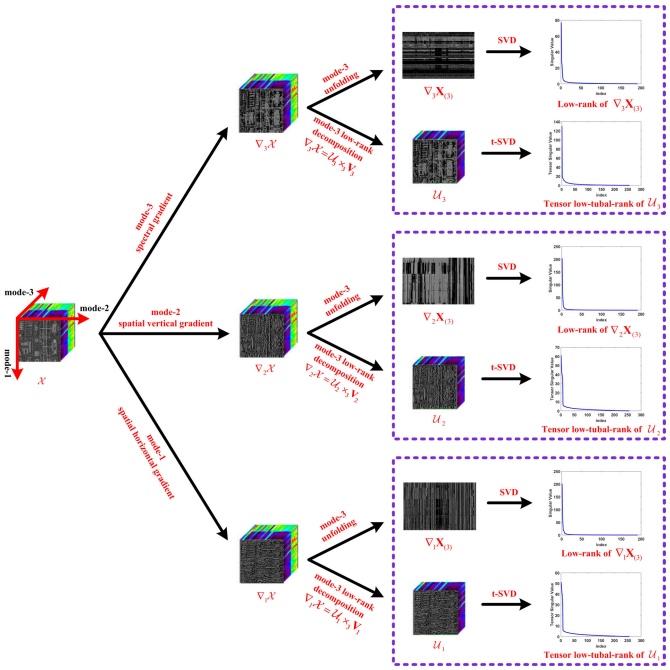


Fig. 1. Gradient subspace-regularized RCT low-tubal-rank prior modeling of HSI $\mathcal{X}$.

Furthermore, we consider its spatial and spectral gradients $\nabla_i \mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $i = 1, 2, 3$, i.e., the spatial horizontal gradient $\nabla_1 \mathcal{X}$, the spatial vertical gradient $\nabla_2 \mathcal{X}$, and the spectral gradient $\nabla_3 \mathcal{X}$, which are defined as

$$\begin{cases} \nabla_1 \mathcal{X}(i_1, i_2, i_3) = \mathcal{X}(i_1 + 1, i_2, i_3) - \mathcal{X}(i_1, i_2, i_3) \\ \nabla_2 \mathcal{X}(i_1, i_2, i_3) = \mathcal{X}(i_1, i_2 + 1, i_3) - \mathcal{X}(i_1, i_2, i_3) \\ \nabla_3 \mathcal{X}(i_1, i_2, i_3) = \mathcal{X}(i_1, i_2, i_3 + 1) - \mathcal{X}(i_1, i_2, i_3) \end{cases} \tag{3}$$

where ∇_1, ∇_2, and ∇_3 are the spatial horizontal gradient operator, the spatial vertical gradient operator, and the spectral gradient operator, respectively.

Remark 1: Specifically, in this article, we use the zero-padding boundary condition for HSI $\mathcal{X}$ to compute the spatial horizontal gradient $\nabla_1 \mathcal{X}$, the spatial vertical gradient $\nabla_2 \mathcal{X}$, and the spectral gradient $\nabla_3 \mathcal{X}$. Since the gradient operators ∇_1, ∇_2, and ∇_3 are all linear operators, which are all full rank under the zero-padding boundary condition for HSI, the gradients $\nabla_1 \mathcal{X}$, $\nabla_2 \mathcal{X}$ and $\nabla_3 \mathcal{X}$ also inherit the low-rank properties as $\mathcal{X}$ in the spectral mode, i.e., $\text{unfold}_3(\nabla_1 \mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$ (or $\nabla_1 \mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$), $\text{unfold}_3(\nabla_2 \mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$ (or $\nabla_2 \mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$), and $\text{unfold}_3(\nabla_3 \mathcal{X}) \in \mathbb{R}^{n_3 \times n_1 n_2}$ (or $\nabla_3 \mathbf{X}_{(3)} \in \mathbb{R}^{n_3 \times n_1 n_2}$) are low rank, which are clearly displayed in Fig. 1. Moreover, we can also have that $\text{rank}(\nabla_1 \mathbf{X}_{(3)}) = \text{rank}(\nabla_2 \mathbf{X}_{(3)}) = \text{rank}(\nabla_3 \mathbf{X}_{(3)}) = \text{rank}(\mathbf{X}_{(3)}) = r \ll n_3$.

Thus, based on the desirable mode-3 low-rank properties of the gradients $\nabla_1 \mathcal{X}$, $\nabla_2 \mathcal{X}$, and $\nabla_3 \mathcal{X}$ in the spectral mode, we propose to apply the mode-3 low-rank decomposition of $\nabla_1 \mathcal{X}$, $\nabla_2 \mathcal{X}$, and $\nabla_3 \mathcal{X}$ into the gradient subspace, and

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clearly obtain that

$$\nabla_i \mathcal{X} = \mathcal{U}_i \times_3 \mathbf{V}_i, \quad i = 1, 2, 3 \tag{4}$$

where $\mathcal{U}_i \in \mathbb{R}^{n_1 \times n_2 \times r}$ denotes the gradient subspace-based RCT, and $\mathbf{V}_i \in \mathbb{R}^{n_3 \times r}$ consisting of r orthogonal bases satisfies $\mathbf{V}_i^{\mathrm{T}}\mathbf{V}_i = \mathbf{I} \in \mathbb{R}^{r \times r}$ and $\mathrm{rank}(\mathbf{V}_i) = r$, $r \ll n_3$, and $\times_3$ denotes the mode-3 product of tensor and matrix.

More specifically, based on the desirable gradient subspace-based mode-3 low-rank decomposition $\nabla_i \mathcal{X} = \mathcal{U}_i \times_3 \mathbf{V}_i$, $i = 1, 2, 3$, in this article, we will particularly focus on investigating and exploring the underlying properties of the gradient subspace-based RCT $\mathcal{U}_i$.

To this end, for the gradient subspace-based RCT $\mathcal{U}_i \in \mathbb{R}^{n_1 \times n_2 \times r}$, we apply the tensor singular value decomposition (t-SVD) of $\mathcal{U}_i$ and clearly obtain the tensor low-tubal-rank property of $\mathcal{U}_i$ (see Fig. 1), i.e., the tensor tubal rank $\mathrm{rank}_{\mathrm{t}}(\mathcal{U}_i)$ is low, which is defined as

$$\mathrm{rank}_{\mathrm{t}}(\mathcal{U}_i) = \#\{m, \mathcal{D}_i(m, m, :) \neq \mathbf{0}\} \tag{5}$$

where $\mathcal{U}_i = \mathcal{C}_i * \mathcal{D}_i * \mathcal{E}_i^{\mathrm{T}}$ is the t-SVD of $\mathcal{X}$, and $\#\{\cdot\}$ is the count operator.

Thus, based on the gradient subspace-based mode-3 low-rank decomposition $\nabla_i \mathcal{X} = \mathcal{U}_i \times_3 \mathbf{V}_i$ and $\mathbf{V}_i^{\mathrm{T}}\mathbf{V}_i = \mathbf{I}$, the unified tensor low-tubal-rank prior term of the gradient subspace-based RCT $\mathcal{U}_i$ is proposed as

$$\mathcal{R}(\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3) = \sum_{i=1}^{3} \mathrm{rank}_{\mathrm{t}}(\mathcal{U}_i). \tag{6}$$

Actually, $\mathrm{rank}_{\mathrm{t}}(\mathcal{U}_i)$ is nonconvex, and directly minimizing the $\mathrm{rank}_{\mathrm{t}}(\mathcal{U}_i)$ is an NP-hard problem. Therefore, we apply the effective LogTNN-based [55] nonconvex tensor low-rank approximation of $\mathrm{rank}_{\mathrm{t}}(\mathcal{U}_i)$ and propose the LogTNN-based nonconvex tensor low-rank prior term of the gradient subspace-based RCT $\mathcal{U}_i$ as

$$\mathcal{R}(\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3) = \sum_{i=1}^{3} \|\mathcal{U}_i\|_{\mathrm{LogTNN}} \tag{7}$$

where the LogTNN of $\mathcal{U}_i$ is defined as $\|\mathcal{U}_i\|_{\mathrm{LogTNN}} = (1/n_3)\sum_{k=1}^{n_3}\sum_{m=1}^{\mathrm{rank}_{\mathrm{t}}(\mathcal{U}_i)} \log(\overline{\mathcal{D}_i}(m, m, k) + \epsilon)$, and $\mathcal{U}_i = \mathcal{C}_i * \mathcal{D}_i * \mathcal{E}_i^{\mathrm{T}}$ denotes the t-SVD of $\mathcal{U}_i$, $\overline{\mathcal{D}_i} = \mathrm{fft}(\mathcal{D}_i, [\,], 3)$, and $\epsilon > 0$ denotes a small parameter.

C. HSI and Stripe-Coupled Nonconvex Tensor Low-Rank Prior

In this section, we will focus on modeling the structural prior of the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ by treating the stripe noise separately.

As we clearly know, the stripe noise $\mathcal{B}$ has strong structural property and low-rank property in fact [46], [47], [48], [49], [50], [51], [52]. Moreover, the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is surely an order-3 tensor as the HSI tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. However, to the best of authors' knowledge, in most of the previous works, the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ either was mainly treated as also the sparse noise $\mathcal{S} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and modeled by the l_1-norm-based sparse prior (i.e., $\|\mathcal{S}\|_1$) or was separated from the sparse noise and simply modeled by the band-by-band matrix low-rank prior (i.e., $\sum_{i_3=1}^{n_3} \mathrm{rank}(\mathcal{B}(:, :, i_3))$ or $\sum_{i_3=1}^{n_3} \|\mathcal{B}(:, :, i_3)\|_*$), which seriously ignore the tensor-based low-rankness prior modeling of the stripe noise $\mathcal{B}$. Moreover, the joint tensor-based low-rankness prior modeling of the HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ and the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ in a unified prior term is not investigated recently.

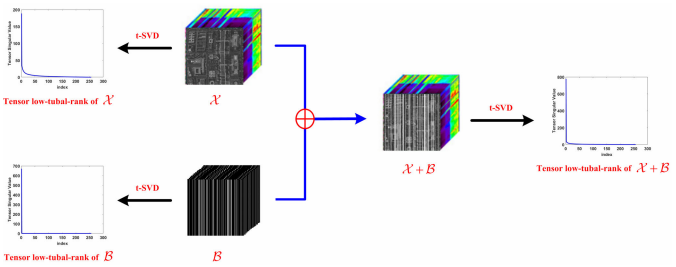


Fig. 2. HSI and stripe-coupled tensor low-tubal-rank prior modeling of $\mathcal{X}+\mathcal{B}$.

To this end, based on the strong structural and low-rank properties of the stripe noise $\mathcal{B}$, we not only treat the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ to be an order-3 tensor as the HSI tensor $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, but also particularly propose to investigate the HSI-guided stripe-coupled tensor low-rank property of stripe noise $\mathcal{B}$ and HSI $\mathcal{X}$ in this article, which can simultaneously exploit the tensor low-rank property of stripe noise $\mathcal{B}$ and HSI $\mathcal{X}$ in a unified prior term.

Thus, for the HSI $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we apply the t-SVD of $\mathcal{X}$, and obviously obtain the tensor low-tubal-rank property of $\mathcal{X}$ (see Fig. 2), i.e., the tensor tubal rank $\mathrm{rank}_{\mathrm{t}}(\mathcal{X})$ is low. Furthermore, the tensor low-tubal-rank prior term of HSI $\mathcal{X}$ is proposed as

$$\mathcal{R}(\mathcal{X}) = \mathrm{rank}_{\mathrm{t}}(\mathcal{X}). \tag{8}$$

Moreover, for the stripe noise $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, we also apply the t-SVD of $\mathcal{B}$, and obviously obtain the tensor low-tubal-rank property of $\mathcal{B}$ (see Fig. 2), i.e., the tensor tubal rank $\mathrm{rank}_{\mathrm{t}}(\mathcal{B})$ is low. Furthermore, the tensor low-tubal-rank prior term of stripe noise $\mathcal{B}$ is proposed as

$$\mathcal{R}(\mathcal{B}) = \mathrm{rank}_{\mathrm{t}}(\mathcal{B}). \tag{9}$$

More specifically, we propose to investigate the HSI-guided stripe-coupled tensor low-rank property of stripe noise $\mathcal{B}$ and HSI $\mathcal{X}$ in a unified prior term, which can exploit the tensor low-rank property of stripe noise $\mathcal{B}$ and HSI $\mathcal{X}$ simultaneously. Thus, we particularly consider the HSI plus stripe (i.e., $\mathcal{X}+\mathcal{B}$) and also apply the t-SVD of $\mathcal{X}+\mathcal{B}$, and obviously obtain that $\mathcal{X}+\mathcal{B}$ has also the tensor low-tubal-rank property (see Fig. 2), i.e., the tensor tubal rank $\mathrm{rank}_{\mathrm{t}}(\mathcal{X}+\mathcal{B})$ is low. Thus, the unified tensor low-tubal-rank prior term of HSI plus stripe $\mathcal{X}+\mathcal{B}$ is proposed as

$$\mathcal{R}(\mathcal{X}, \mathcal{B}) = \mathrm{rank}_{\mathrm{t}}(\mathcal{X} + \mathcal{B}). \tag{10}$$

Moreover, based on the nonconvexity of $\mathrm{rank}_{\mathrm{t}}(\mathcal{X}+\mathcal{B})$ and the NP-hard property of directly minimizing $\mathrm{rank}_{\mathrm{t}}(\mathcal{X}+\mathcal{B})$, we also apply the effective LogTNN-based nonconvex tensor low-rank approximation of $\mathrm{rank}_{\mathrm{t}}(\mathcal{X}+\mathcal{B})$ and propose the

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unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term as

$$\mathcal{R}(\mathcal{X},\mathcal{B}) = \|\mathcal{X}+\mathcal{B}\|_{\text{LogTNN}}. \tag{11}$$

D. Proposed GSHSNTLR Model

Therefore, by combining the LogTNN-based nonconvex tensor low-rank prior term of the gradient subspace-based RCTs in (7) and the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term in (11), we propose a novel and unified HSI destriping and denoising model with GSHSNTLR, which is constructed as

$$\begin{aligned}(\hat{\mathcal{X}},\hat{\mathcal{S}},\hat{\mathcal{B}},\hat{\mathcal{U}}_i,\hat{\mathbf{V}}_i) = \arg\min_{\mathcal{X},\mathcal{S},\mathcal{B},\mathcal{U}_i,\mathbf{V}_i} &\frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2+\beta\|\mathcal{S}\|_1 \\ &+\alpha\sum_{i=1}^{3}\|\mathcal{U}_i\|_{\text{LogTNN}}+\lambda\|\mathcal{X}+\mathcal{B}\|_{\text{LogTNN}} \\ \text{s.t.}\quad &\nabla_i\mathcal{X}=\mathcal{U}_i\times_3\mathbf{V}_i \\ &\mathbf{V}_i^T\mathbf{V}_i=\mathbf{I},\quad i=1,2,3\end{aligned} \tag{12}$$

where $(1/2)\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2$ models the Gaussian noise tensor $\mathcal{G}$, $\|\mathcal{S}\|_1$ models the sparse noise tensor $\mathcal{S}$, and β, α, and λ are the regularization parameters.

III. Optimization Algorithm

In this section, we solve the proposed GSHSNTLR model (12) by utilizing the ADMM method.

To this end, we introduce an auxiliary tensor variable $\mathcal{A}\in\mathbb{R}^{n_1\times n_2\times n_3}$ such that $\mathcal{A}=\mathcal{X}+\mathcal{B}$ and formulate the proposed GSHSNTLR model (12) equivalently as

$$\begin{aligned}\min_{\mathcal{X},\mathcal{S},\mathcal{B},\mathcal{A},\mathcal{U}_i,\mathbf{V}_i} &\frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2+\beta\|\mathcal{S}\|_1 \\ &+\alpha\sum_{i=1}^{3}\|\mathcal{U}_i\|_{\text{LogTNN}}+\lambda\|\mathcal{A}\|_{\text{LogTNN}} \\ \text{s.t.}\quad &\nabla_i\mathcal{X}=\mathcal{U}_i\times_3\mathbf{V}_i \\ &\mathbf{V}_i^T\mathbf{V}_i=\mathbf{I},\quad i=1,2,3 \\ &\mathcal{A}=\mathcal{X}+\mathcal{B}.\end{aligned} \tag{13}$$

Then, we can formulate the augmented Lagrangian function of model (13) as

$$\begin{aligned}f(\mathcal{X},\mathcal{S},\mathcal{B},\mathcal{A},\mathcal{U}_i,\mathbf{V}_i,\mathcal{H},\mathcal{P}_i) = &\frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2+\beta\|\mathcal{S}\|_1 \\ &+\sum_{i=1}^{3}\left(\alpha\|\mathcal{U}_i\|_{\text{LogTNN}}+\frac{\eta}{2}\|\mathcal{U}_i\times_3\mathbf{V}_i-\nabla_i\mathcal{X}-\mathcal{P}_i\|_F^2\right) \\ &+\lambda\|\mathcal{A}\|_{\text{LogTNN}}+\frac{\mu}{2}\|\mathcal{A}-(\mathcal{X}+\mathcal{B})-\mathcal{H}\|_F^2\end{aligned} \tag{14}$$

where $\mathcal{H}\in\mathbb{R}^{n_1\times n_2\times n_3}$ and $\mathcal{P}_i\in\mathbb{R}^{n_1\times n_2\times n_3}$ $(i=1,2,3)$ stand for the Lagrange multiplier tensors, and μ and η denote the penalty parameters.

Given initial $\mathcal{X}^{(0)}$, $\mathcal{S}^{(0)}$, $\mathcal{B}^{(0)}$, $\mathcal{A}^{(0)}$, $\mathcal{H}^{(0)}$, $\mathcal{U}_i^{(0)}$, $\mathbf{V}_i^{(0)}$, and $\mathcal{P}_i^{(0)}$ $(i=1,2,3)$, we can alternatively and iteratively optimize the model (14) at $(k+1)$th iteration by the following optimization subproblems.

A. Optimization of $\mathcal{S}$

From the model (14), we can derive the $\mathcal{S}^{(k+1)}$ subproblem as

$$\mathcal{S}^{(k+1)} = \arg\min_{\mathcal{S}}\frac{1}{2}\|\mathcal{Y}-\mathcal{X}^{(k)}-\mathcal{S}-\mathcal{B}^{(k)}\|_F^2+\beta\|\mathcal{S}\|_1. \tag{15}$$

Thus, we use the soft-thresholding method to solve $\mathcal{S}^{(k+1)}$ in the closed form as

$$\mathcal{S}^{(k+1)} = \mathfrak{R}_\beta\left(\mathcal{Y}-\mathcal{X}^{(k)}-\mathcal{B}^{(k)}\right) \tag{16}$$

where the soft-thresholding operator $\mathfrak{R}_\beta(\cdot)$ is defined as

$$\mathfrak{R}_\beta(\mathcal{Z}) = \begin{cases}\mathcal{Z}-\beta, & \text{if } \mathcal{Z}>\beta \\ \mathcal{Z}+\beta, & \text{if } \mathcal{Z}<-\beta \\ 0, & \text{otherwise.}\end{cases} \tag{17}$$

B. Optimization of $\mathcal{B}$

From the model (14), we can derive the $\mathcal{B}^{(k+1)}$ subproblem as

$$\begin{aligned}\mathcal{B}^{(k+1)} = \arg\min_{\mathcal{B}} &\frac{\mu}{2}\|\mathcal{A}^{(k)}-\left(\mathcal{X}^{(k)}+\mathcal{B}\right)-\mathcal{H}^{(k)}\|_F^2 \\ &+\frac{1}{2}\|\mathcal{Y}-\mathcal{X}^{(k)}-\mathcal{S}^{(k+1)}-\mathcal{B}\|_F^2\end{aligned} \tag{18}$$

which is obviously solved as

$$\mathcal{B}^{(k+1)} = \frac{\mu\left(\mathcal{A}^{(k)}-\mathcal{X}^{(k)}-\mathcal{H}^{(k)}\right)+\mathcal{Y}-\mathcal{X}^{(k)}-\mathcal{S}^{(k+1)}}{\mu+1}. \tag{19}$$

C. Optimization of $\mathcal{A}$

From the model (14), we can derive the $\mathcal{A}^{(k+1)}$ subproblem as

$$\mathcal{A}^{(k+1)} = \arg\min_{\mathcal{A}}\frac{\mu}{2}\|\mathcal{A}-\left(\mathcal{X}^{(k)}+\mathcal{B}^{(k+1)}\right)-\mathcal{H}^{(k)}\|_F^2+\lambda\|\mathcal{A}\|_{\text{LogTNN}} \tag{20}$$

which can be clearly solved by the tensor singular value thresholding (t-SVT) method [55] as

$$\mathcal{A}^{(k+1)} = \text{prox}_{\lambda/\mu}^{\epsilon}\left(\mathcal{X}^{(k)}+\mathcal{B}^{(k+1)}+\mathcal{H}^{(k)}\right) \tag{21}$$

where the t-SVT operator $\text{prox}_\tau^\epsilon(\cdot)$ is defined as

$$\text{prox}_\tau^\epsilon(\mathcal{Z}) = \mathcal{C}*\mathcal{D}_\tau^\epsilon*\mathcal{E}^{\text{T}} \tag{22}$$

where $\mathcal{Z}=\mathcal{C}*\mathcal{D}*\mathcal{E}^{\text{T}}$ is the t-SVD of $\mathcal{Z}$, $\bar{\mathcal{D}}=\text{fft}(\mathcal{D},[\,],3)$, $\mathcal{D}_\tau^\epsilon=\text{ifft}(\bar{\mathcal{D}}_\tau^\epsilon,[\,],3)$, and $\bar{\mathcal{D}}_\tau^\epsilon(m,n,k)$ satisfies that

$$\bar{\mathcal{D}}_\tau^\epsilon(m,n,k) = \begin{cases}0, & \text{if } c_2\le 0 \\ \dfrac{c_1+\sqrt{c_2}}{2}, & \text{if } c_2>0\end{cases} \tag{23}$$

where $c_1=|\bar{\mathcal{D}}(m,n,k)|-\epsilon$ and $c_2=c_1^2-4(\tau-\epsilon|\bar{\mathcal{D}}(m,n,k)|)$.

D. Optimization of $\mathcal{U}_i$

From the model (14), we can derive the $\mathcal{U}_i^{(k+1)}$ $(i=1,2,3)$ subproblem as

$$\mathcal{U}_i^{(k+1)} = \arg\min_{\mathcal{U}_i}\frac{\eta}{2}\|\mathcal{U}_i\times_3\mathbf{V}_i^{(k)}-\nabla_i\mathcal{X}^{(k)}-\mathcal{P}_i^{(k)}\|_F^2+\alpha\|\mathcal{U}_i\|_{\text{LogTNN}} \tag{24}$$

which can also be solved by the t-SVT method [55] as

$$\mathcal{U}_i^{(k+1)} = \text{prox}_{\alpha/\eta}^{\epsilon}\left(\left(\nabla_i\mathcal{X}^{(k)}+\mathcal{P}_i^{(k)}\right)\times_3\mathbf{V}_i^{(k)^{\text{T}}}\right). \tag{25}$$

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E. Optimization of $\mathbf{V}_i$

Then, the $\mathbf{V}_i^{(k+1)}$ $(i = 1, 2, 3)$ subproblem is formulated as

$$\mathbf{V}_i^{(k+1)} = \underset{\mathbf{V}_i, \mathbf{V}_i^T\mathbf{V}_i=\mathbf{I}}{\arg\min} \frac{\eta}{2}\|\mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i - \nabla_i\mathcal{X}^{(k)} - \mathcal{P}_i^{(k)}\|_F^2. \quad (26)$$

Specifically, based on the definition and property of mode-3 product of tensor and matrix, namely, $\nabla_i\mathcal{X} = \mathcal{U}_i \times_3 \mathbf{V}_i$ means that $\text{unfold}_3(\nabla_i\mathcal{X}) = \mathbf{V}_i\,\text{unfold}_3(\mathcal{U}_i)$, then we can equivalently derive the $\mathbf{V}_i^{(k+1)}$ subproblem (26) as

$$\mathbf{V}_i^{(k+1)} = \underset{\mathbf{V}_i, \mathbf{V}_i^T\mathbf{V}_i=\mathbf{I}}{\arg\max} \langle \text{unfold}_3\left(\nabla_i\mathcal{X}^{(k)} + \mathcal{P}_i^{(k)}\right)\left(\text{unfold}_3\left(\mathcal{U}_i^{(k+1)}\right)\right)^{\mathrm{T}}, \mathbf{V}_i\rangle \quad (27)$$

where $\text{unfold}_3(\nabla_i\mathcal{X}^{(k)} + \mathcal{P}_i^{(k)}) \in \mathbb{R}^{n_3\times n_1 n_2}$ denotes the mode-3 unfolding of $\nabla_i\mathcal{X}^{(k)} + \mathcal{P}_i^{(k)}$, $\text{unfold}_3(\mathcal{U}_i^{(k+1)}) \in \mathbb{R}^{r\times n_1 n_2}$ denotes the mode-3 unfolding of $\mathcal{U}_i^{(k+1)}$, $\langle \mathbf{W}_1, \mathbf{W}_2\rangle = \text{tr}(\mathbf{W}_1^{\mathrm{T}}\mathbf{W}_2)$ denotes the inner product of matrices $\mathbf{W}_1 \in \mathbb{R}^{n_3\times r}$ and $\mathbf{W}_2 \in \mathbb{R}^{n_3\times r}$, and $\text{tr}(\cdot)$ denotes the trace operator of the matrix.

By using singular value decomposition (SVD) and [56, Th. 1], we can further solve $\mathbf{V}_i^{(k+1)}$ as

$$\mathbf{V}_i^{(k+1)} = \mathbf{C}_i^{(k)}\,\mathbf{E}_i^{(k)^{\mathrm{T}}} \quad (28)$$

where the matrices $\mathbf{C}_i^{(k)}$ and $\mathbf{E}_i^{(k)}$ are both obtained by the SVD of $\text{unfold}_3(\nabla_i\mathcal{X}^{(k)} + \mathcal{P}_i^{(k)})(\text{unfold}_3(\mathcal{U}_i^{(k+1)}))^{\mathrm{T}}$, i.e.,

$$\text{unfold}_3\left(\nabla_i\mathcal{X}^{(k)} + \mathcal{P}_i^{(k)}\right)\left(\text{unfold}_3\left(\mathcal{U}_i^{(k+1)}\right)\right)^{\mathrm{T}} = \mathbf{C}_i^{(k)}\mathbf{D}_i^{(k)}\mathbf{E}_i^{(k)^{\mathrm{T}}}. \quad (29)$$

F. Optimization of $\mathcal{X}$

From the model (14), we can derive the $\mathcal{X}^{(k+1)}$ subproblem as

$$\begin{aligned}\mathcal{X}^{(k+1)} = \underset{\mathcal{X}}{\arg\min}\ & \frac{1}{2}\|\mathcal{Y} - \mathcal{X} - \mathcal{S}^{(k+1)} - \mathcal{B}^{(k+1)}\|_F^2 \\ & + \frac{\mu}{2}\|\mathcal{A}^{(k+1)} - \left(\mathcal{X} + \mathcal{B}^{(k+1)}\right) - \mathcal{H}^{(k)}\|_F^2 \\ & + \frac{\eta}{2}\sum_{i=1}^{3}\|\mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i^{(k+1)} - \nabla_i\mathcal{X} - \mathcal{P}_i^{(k)}\|_F^2. \quad (30)\end{aligned}$$

Obviously, it is solved by the below linear system, i.e.,

$$\begin{aligned}\left(\mathcal{I} + \mu\mathcal{I} + \eta\sum_{i=1}^{3}\nabla_i^{\mathrm{T}}\nabla_i\right)\mathcal{X} = {} & \mathcal{Y} - \mathcal{S}^{(k+1)} - \mathcal{B}^{(k+1)} \\ & + \mu\left(\mathcal{A}^{(k+1)} - \mathcal{B}^{(k+1)} - \mathcal{H}^{(k)}\right) \\ & + \eta\sum_{i=1}^{3}\nabla_i^{\mathrm{T}}\left(\mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i^{(k+1)} - \mathcal{P}_i^{(k)}\right). \quad (31)\end{aligned}$$

Thus, we use the fast Fourier transform (FFT) to solve (31) efficiently and obtain $\mathcal{X}^{(k+1)}$ as

$$\mathcal{X}^{(k+1)} = \text{ifftn}\left(\frac{\text{fftn}(\mathcal{M} + \mu\mathcal{N} + \eta\mathcal{Q})}{1 + \mu + \eta\sum_{i=1}^{3}|\text{fftn}(\nabla_i)|^2}\right) \quad (32)$$

Algorithm 1 GSHSNTLR

Input: Noisy HSI $\mathcal{Y} \in \mathbb{R}^{n_1\times n_2\times n_3}$, parameters β, α, η, λ, μ, *MaxIter*.

Initialization: $k = 0$, $\mathcal{X}^{(0)} = \mathcal{Y}$, $\mathcal{S}^{(0)} = \mathcal{B}^{(0)} = \mathcal{A}^{(0)} = \mathcal{H}^{(0)} = \mathcal{U}_i^{(0)} = \mathcal{P}_i^{(0)} = \mathcal{O}$, $\mathbf{V}_i^{(0)} = \mathbf{O}$, $i = 1, 2, 3$, where $\mathcal{O} \in \mathbb{R}^{n_1\times n_2\times n_3}$ denotes the zero tensor, $\mathbf{O} \in \mathbb{R}^{n_3\times r}$ denotes the zero matrix.

Iterations:

For $k = 0$ **to** *MaxIter* $- 1$ **Do**
  Solve $\mathcal{S}^{(k+1)}$ via (16)
  Solve $\mathcal{B}^{(k+1)}$ via (19)
  Solve $\mathcal{A}^{(k+1)}$ via (21)
  Solve $\mathcal{U}_i^{(k+1)}$ via (25)
  Solve $\mathbf{V}_i^{(k+1)}$ via (26)-(29)
  Solve $\mathcal{X}^{(k+1)}$ via (32)
  Update $\mathcal{H}^{(k+1)}$ and $\mathcal{P}_i^{(k+1)}$ via (33)
End For

Output: Restored HSI $\mathcal{X}^{(MaxIter)}$.

where $\mathcal{M} = \mathcal{Y} - \mathcal{S}^{(k+1)} - \mathcal{B}^{(k+1)}$, $\mathcal{N} = \mathcal{A}^{(k+1)} - \mathcal{B}^{(k+1)} - \mathcal{H}^{(k)}$, $\mathcal{Q} = \sum_{i=1}^{3}\nabla_i^{\mathrm{T}}(\mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i^{(k+1)} - \mathcal{P}_i^{(k)})$, $\text{fftn}(\cdot)$ is the 3-D Fourier transform operator, and $\text{ifftn}(\cdot)$ is the inverse 3-D Fourier transform operator.

G. Update $\mathcal{H}$ and $\mathcal{P}_i$

Finally, we update the Lagrangian multiplier tensors $\mathcal{H}^{(k+1)}$ and $\mathcal{P}_i^{(k+1)}$ $(i = 1, 2, 3)$ as

$$\begin{cases}\mathcal{H}^{(k+1)} = \mathcal{H}^{(k)} + \mathcal{X}^{(k+1)} + \mathcal{B}^{(k+1)} - \mathcal{A}^{(k+1)} \\ \mathcal{P}_i^{(k+1)} = \mathcal{P}_i^{(k)} + \nabla_i\mathcal{X}^{(k+1)} - \mathcal{U}_i^{(k+1)} \times_3 \mathbf{V}_i^{(k+1)},\ i = 1, 2, 3.\end{cases} \quad (33)$$

Moreover, the concise description of the proposed GSHSNTLR algorithm is summarized in Algorithm 1.

IV. Experiments

A. Experimental Setting

In this section, we detailedly describe the experiments on both simulated and real HSI datasets for comprehensively evaluating the noise removal effectiveness of the proposed GSHSNTLR method. Additionally, we compare the proposed GSHSNTLR method with many HSI denoising methods including low-rank matrix recovery (LRMR) [17], TV-regularized low-rank matrix factorization (LRTV) [33], TV-regularized low-rank tensor decomposition (LRTDTV) [38], 3DTNN [31], group sparsity-regularized low-rank tensor decomposition (LRTDGS) [25], RCTV [15], 3DCTV [43], fast graph Laplacian regularization (FGLR) [51], E-3DTV [14], factor group sparsity-regularized nonconvex low-rank (FGSLR) approximation [21], and fast and parameter-free HSI mixed noise removal method (FastHyMix) [9].

In the simulated denoising experiments, we use the Pavia City Center (PaviaC) and Washington DC (WDC) mall datasets to simulate different types of noisy HSI for experimental testing. Specifically, various types of noise, including

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Gaussian noise, impulse noise, deadlines, and stripe noise, are added to the clean HSI to simulate and generate the noisy HSI. Furthermore, we use the mean peak signal-to-noise ratio (MPSNR), mean structural similarity (MSSIM), and spectral angle mapper (SAM) for quantitative assessments of the simulated denoising experiments.

By contrast, in the real denoising experiments, we use the real GaoFen-5 (GF-5) Shanghai dataset, the real Indian Pines dataset, and the real HYDICE Urban dataset directly for experimental testing.

Moreover, in our experiments, we normalize the intensity values of each spectral band of HSI to the range of [0, 1].

Besides, we implement all the compared HSI denoising methods by applying MATLAB(R2018b) on a personal computer configured with Windows 10 and 2.20-GHz Intel[1] Core[2] i7-8750H CPU and 16-GB RAM in our experiments.

B. Denoising Experiments on Simulated HSI Datasets

1) Simulation Setting: Specifically, the subimage of PaviaC dataset (size: $200 \times 200 \times 80$) and the subimage of WDC dataset (size: $256 \times 256 \times 191$) are adopted for the denoising experiments on simulated HSI datasets.

In detail, we consider to simulate and add the following four cases of noise for the denoising experiments on PaviaC and WDC datasets, which are listed as follows.

Case 1 (Gaussian + Stripe): First, the Gaussian noise with the standard deviation of the range of $[0.1, 0.2]$ is randomly added to each spectral band of HSI. Then, the stripe noise is randomly added to 40% of spectral bands of HSI and 30% of the columns of these bands.

Case 2 (Gaussian + Stripe + Impulse): Based on the same Gaussian noise and stripe noise added in Case 1, we also randomly add the impulse noise of density of 1% to 20% spectral bands of HSI.

Case 3 (Gaussian + Stripe + Deadlines): Based on the same Gaussian noise and stripe noise added in Case 1, we randomly select 11–30 bands with the stripe noise to further add the deadlines, where the number of deadlines is chosen from the range of $[1, 5]$, and the width of the deadlines is chosen from the range of $[1, 3]$.

Case 4 (Gaussian + Stripe + Impulse + Deadlines): First, we add the same Gaussian noise, stripe noise, and deadlines to the HSI based on Case 3. Then, we also add the same impulse noise based on Case 2 to finally simulate the noisy HSI.

Case 5 (Gaussian + Wide Vertical Stripe + Impulse + Deadlines): First, we add the same Gaussian noise, impulse noise, and deadlines to the HSI based on Case 4. Then, we also add the wide vertical stripe noise to the continuous 60 columns of the HSI to finally simulate the noisy HSI.

2) Quantitative Assessments: Then, the detailed quantitative and numerical MPSNR, MSSIM, and SAM values for assessing these denoising methods on simulated PaviaC and WDC datasets are shown in Table I, in which the best values in each case are bolded, and the second-best values are underlined. Obviously, the proposed GSHSNTLR method performs the best MPSNR, MSSIM, and SAM values for all the cases compared to other methods, i.e., the best quantitative denoising performance for removing the stripe noise and other mixed noise.

3) Visual Assessments: Moreover, for more complete visual comparisons and assessments in the aspects of restored HSI, residual images between the restored HSI and original HSI, and the fitting of spectral curves between the restored HSI and original HSI, we particularly display the visual denoising results of all the compared methods on the simulated PaviaC dataset and WDC dataset under Case 4 and Case 5, which are two of the most representative cases with serious mixed noise to validate the denoising capacity of different methods to remove the stripe noise and other mixed noise.

To this end, Figs. 3 and 4 display the visual denoising results of restored images and residual images on 80th band of the simulated PaviaC dataset under Case 4, respectively, Fig. 5 displays the results of spectral curves on the simulated PaviaC dataset under Case 4 at spatial pixel $(102, 99)$, and Fig. 6 further displays the pseudo-color images of restored images of different methods on the simulated PaviaC dataset under Case 5 to better illustrate and compare the spectral recovery performance of different methods. Besides, Figs. 7 and 8 display the visual denoising results of restored images and residual images on 123rd band of the simulated WDC dataset under Case 4, respectively, Fig. 9 displays the results of spectral curves on the simulated WDC dataset under Case 4 at spatial pixel $(220, 93)$, and Fig. 10 further displays the pseudo-color images of restored images of different methods on the simulated WDC dataset under Case 4 to better illustrate and compare the spectral recovery performance of different methods.

As clearly displayed in Figs. 3 and 7, where the enlarged details of the subregions are also displayed for enhanced visual comparison, LRMR cannot completely reduce the stripe noise or other hybrid noise [see Figs. 3(c) and 7(c)]. LRTV can better reduce the stripe noise and other hybrid noise while oversmoothing the image structures and details [see Figs. 3(d) and 7(d)]. LRTDTV also remains some hybrid noise [see Fig. 3(e)] and slightly smooths fewer image structures and details [see Fig. 7(e)]. In particular, 3DTNN cannot completely remove the stripe noise [see Figs. 3(f) and 7(f)]. LRTDGS cannot completely remove the hybrid noise, especially for the stripe noise [see Figs. 3(g) and 7(g)]. In general, E-3DTV, RCTV, and 3DCTV better reduce the hybrid noise and preserve image structures sharper [see Figs. 3(h), (i), and (k) and 7(h), (i), and (k)]. FGSLR cannot completely remove the hybrid noise, especially for the stripe noise [see Figs. 3(j) and 7(j)]. FGLR also remains some hybrid noise to some extent [see Figs. 3(l) and 7(l)]. FastHyMix cannot completely remove the hybrid noise, especially for the stripe noise [see Figs. 3(m) and 7(m)]. More obviously, among all the compared approaches, the proposed GSHSNTLR approach can best reduce the hybrid noise and preserve image structures and details sharpest [see Figs. 3(n) and 7(n)], which gives the best visual quality of the restored HSI.

Moreover, with the accordingly residual images between the restored HSI and original HSI displayed in Figs. 4 and 8, it is more clear that the proposed GSHSNTLR approach

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TABLE I

NUMERICAL VALUES OF MPSNR, MSSIM, AND SAM FOR THE SIMULATED DENOISING EXPERIMENTS USING THE PAVIAC AND WDC DATASETS

Datasets	Cases	Indexes	Noisy	LRMR	LRTV	LRTDTV	3DTNN	LRTDGS	E-3DTV	RCTV	FGSLR	3DCTV	FGLR	FastHyMix	GSHSNTLR
PaviaC	Case 1	MPSNR	16.41	27.94	26.17	27.75	27.86	28.46	28.37	29.24	28.68	28.41	29.33	**31.12**	30.99
		MSSIM	0.2483	0.8243	0.6898	0.7891	0.7927	0.8311	0.8519	0.8657	0.8316	0.8380	0.8677	0.8933	**0.9120**
		SAM	33.9819	9.4982	12.5832	8.8592	10.0039	11.1027	6.8614	8.6803	11.0045	8.9280	6.6617	8.4920	**5.2337**
	Case 2	MPSNR	16.71	28.10	25.92	28.19	28.06	28.46	28.53	29.19	28.59	28.64	29.48	**31.25**	31.00
		MSSIM	0.2591	0.8258	0.6745	0.7998	0.8071	0.8297	0.8412	0.8631	0.8239	0.8408	0.8692	0.8998	**0.9104**
		SAM	33.184	9.4217	13.2664	8.1435	9.4654	11.110	6.1922	8.6279	11.4091	8.6883	6.5245	8.2659	**5.0449**
	Case 3	MPSNR	16.40	27.77	26.17	27.74	27.38	28.10	28.54	29.05	28.52	28.36	29.18	30.65	**30.90**
		MSSIM	0.2492	0.8156	0.6909	0.7884	0.7642	0.7789	0.8482	0.8593	0.8320	0.8365	0.8636	0.8738	**0.9092**
		SAM	34.0081	9.7319	12.4823	8.6012	11.3569	13.2071	6.4337	8.6116	11.0321	8.6687	6.4191	8.9166	**4.9549**
	Case 4	MPSNR	16.66	28.06	26.34	28.21	27.88	28.17	28.81	29.17	28.62	28.67	29.51	**31.05**	**31.05**
		MSSIM	0.2573	0.8246	0.6862	0.7991	0.7962	0.7860	0.8494	0.8670	0.8250	0.8404	0.8688	0.8892	**0.9100**
		SAM	33.4615	9.4376	12.6064	8.2918	10.0989	12.2843	6.0628	8.3036	11.1884	8.4615	6.3724	8.5018	**4.9131**
	Case 5	MPSNR	16.56	27.54	26.97	27.92	28.15	28.31	27.81	28.66	29.13	28.60	29.20	**30.70**	30.43
		MSSIM	0.2590	0.7893	0.7405	0.7957	0.8296	0.7891	0.8100	0.8416	0.8403	0.8275	0.8611	0.8865	**0.8963**
		SAM	34.0961	9.8803	9.7506	7.2623	7.8573	10.6607	6.6187	7.3843	7.6115	7.2507	6.1678	6.2719	**5.3535**
WDC	Case 1	MPSNR	16.64	28.85	25.12	27.38	27.42	28.76	27.19	29.65	29.73	28.94	27.91	**32.11**	31.75
		MSSIM	0.2676	0.8484	0.6212	0.7715	0.7546	0.8139	0.7833	0.8771	0.8321	0.8351	0.8472	0.9088	**0.9201**
		SAM	36.8730	11.4138	14.9332	10.8673	13.3587	13.3112	10.6816	10.1278	13.5512	11.8671	11.0906	8.5820	**7.7458**
	Case 2	MPSNR	16.67	28.84	25.12	27.40	27.47	28.81	28.01	29.66	29.75	29.10	27.82	31.44	**31.55**
		MSSIM	0.2684	0.8485	0.6232	0.7742	0.7562	0.8156	0.8276	0.8790	0.8371	0.8425	0.8473	0.8951	**0.9155**
		SAM	36.5139	11.2010	14.3950	10.5375	13.1115	13.1292	10.2314	9.7104	12.8680	11.1732	10.7458	9.2474	**8.0462**
	Case 3	MPSNR	16.67	28.98	25.13	27.42	27.54	28.82	27.65	29.75	29.83	29.10	27.83	**32.23**	31.86
		MSSIM	0.2700	0.8552	0.6294	0.7786	0.7591	0.8152	0.8168	0.8865	0.8374	0.8462	0.8497	0.9116	**0.9252**
		SAM	36.8820	11.2562	14.5018	10.5920	13.2110	13.0689	10.4955	9.7539	13.1238	11.2868	11.3665	8.2903	**7.4110**
	Case 4	MPSNR	16.51	28.78	25.19	27.45	27.41	28.65	28.49	29.58	29.61	29.11	27.70	**31.53**	31.43
		MSSIM	0.2616	0.8484	0.6260	0.7734	0.7557	0.8101	0.8451	0.8792	0.8325	0.8409	0.8486	0.8955	**0.9147**
		SAM	36.96	11.3714	14.0638	10.5141	13.5218	13.2485	10.3631	9.7729	12.7533	11.2521	11.1142	9.1605	**8.1668**
	Case 5	MPSNR	16.84	28.49	25.70	27.10	27.79	29.08	27.20	26.79	30.21	29.06	27.20	28.83	**30.52**
		MSSIM	0.2833	0.8223	0.6695	0.7669	0.7974	0.8236	0.7961	0.8229	0.8496	0.8359	0.8411	**0.8967**	0.8837
		SAM	35.9629	12.0318	11.6836	10.8472	12.0195	10.9910	11.9415	12.2915	10.5243	10.9468	11.0812	**8.0889**	9.7337

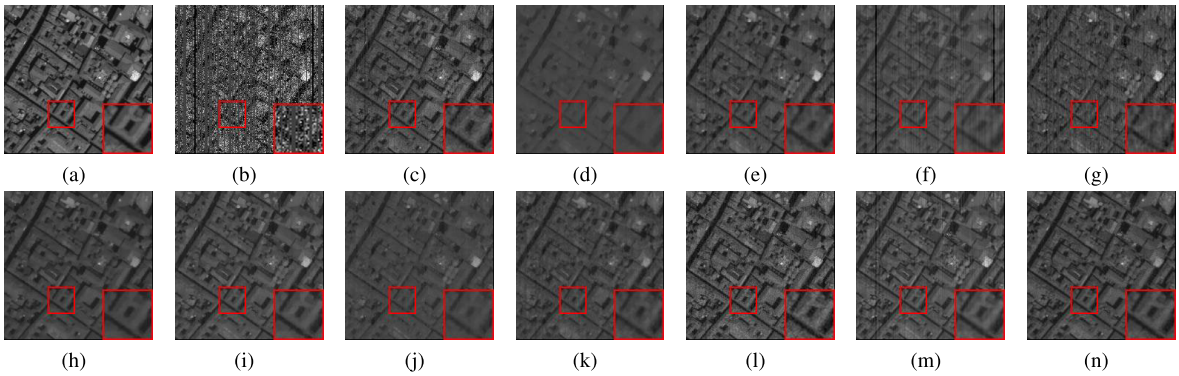


Fig. 3. Restored HSI results of different denoising methods on 80th band of the simulated PaviaC dataset under Case 4. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) 3DTNN. (g) LRTDGS. (h) E-3DTV. (i) RCTV. (j) FGSLR. (k) 3DCTV. (l) FGLR. (m) FastHyMix. (n) GSHSNTLR.



Fig. 4. Residual image results of different denoising methods on 80th band of the simulated PaviaC dataset under Case 4. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) 3DTNN. (g) LRTDGS. (h) E-3DTV. (i) RCTV. (j) FGSLR. (k) 3DCTV. (l) FGLR. (m) FastHyMix. (n) GSHSNTLR.

gives the best performance of residual images with the least residuals, which means that the restored HSI of the proposed GSHSNTLR approach is structurally closest to the original HSI.

Besides, with the accordingly fitting results of spectral curves between the restored HSI and original HSI displayed in Figs. 5 and 9, where the black color denotes the spectral curve of the original HSI and the red color denotes the spectral

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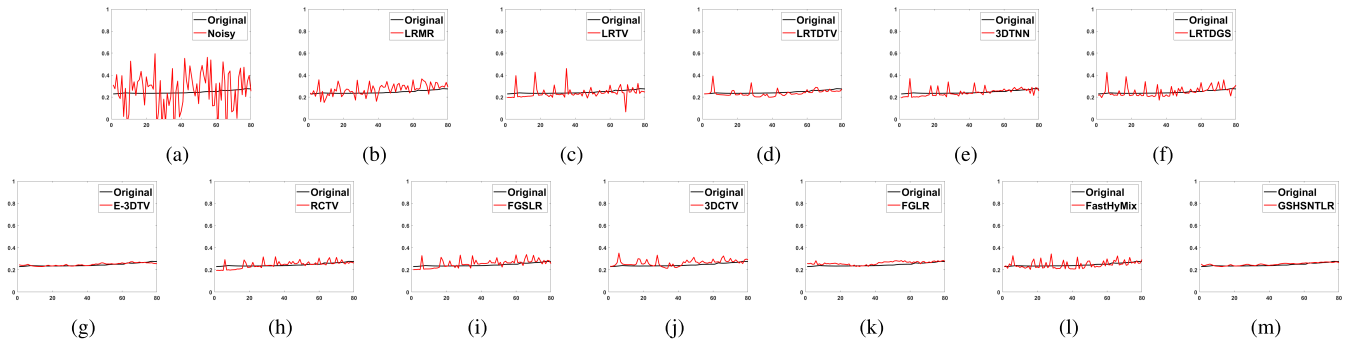


Fig. 5. Fitting results of spectral curves of different denoising methods on the simulated PaviaC dataset under Case 4 at spatial pixel (102, 99). Note that the black color denotes the spectral curve of original HSI, and the red color denotes the spectral curve of restored HSI of different denoising methods, where the closer the red color fits the black color, the better the method preserves the spectral curve. (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

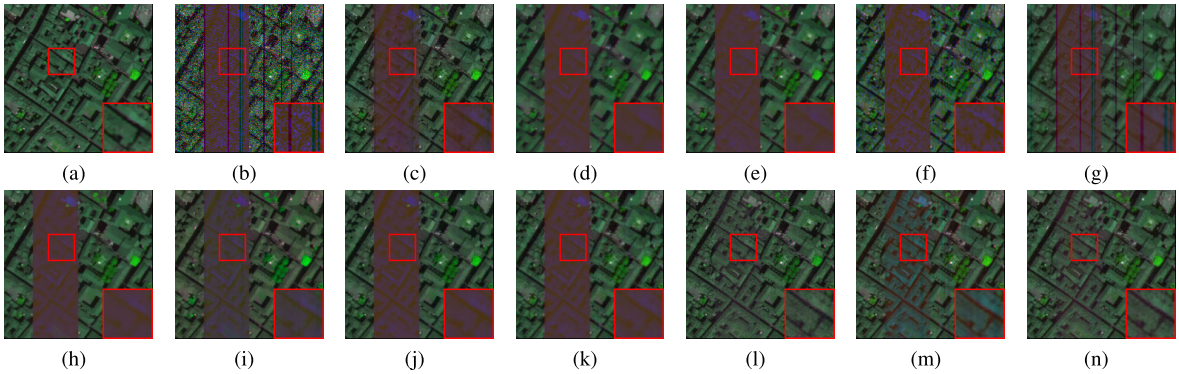


Fig. 6. Pseudo-color images of restored HSI results of different denoising methods on (7, 61, 20) bands of the simulated PaviaC dataset under Case 5. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) 3DTNN. (g) LRTDGS. (h) E-3DTV. (i) RCTV. (j) FGSLR. (k) 3DCTV. (l) FGLR. (m) FastHyMix. (n) GSHSNTLR.

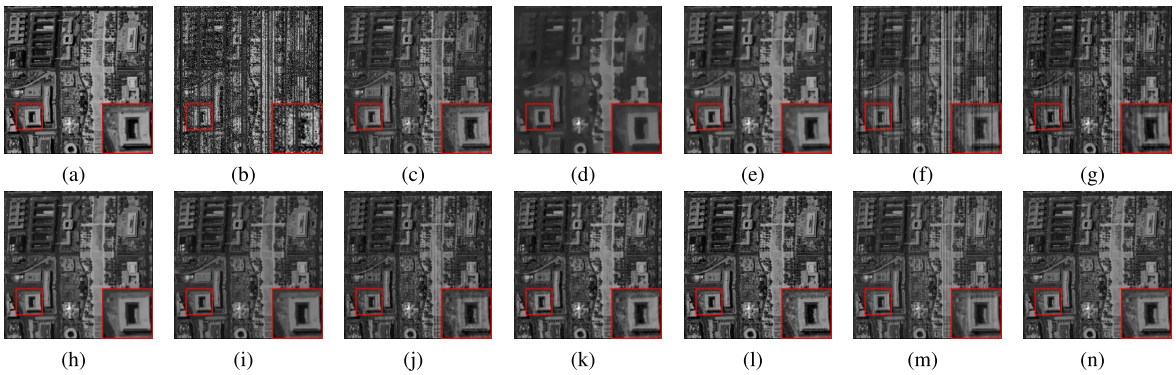


Fig. 7. Restored HSI results of different denoising methods on 123rd band of the simulated WDC dataset under Case 4. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) 3DTNN. (g) LRTDGS. (h) E-3DTV. (i) RCTV. (j) FGSLR. (k) 3DCTV. (l) FGLR. (m) FastHyMix. (n) GSHSNTLR.



Fig. 8. Residual image results of different denoising methods on 123th band of the simulated WDC dataset under Case 4. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) 3DTNN. (g) LRTDGS. (h) E-3DTV. (i) RCTV. (j) FGSLR. (k) 3DCTV. (l) FGLR. (m) FastHyMix. (n) GSHSNTLR.

curve of restored HSI of different denoising methods, where the closer the red color fits the black color, the better the method preserves the spectral curve, it is also more clear that the proposed GSHSNTLR approach gives the best fitting performance of spectral curves with the original HSI, which further validates the superiority of the proposed GSHSNTLR approach.

Meanwhile, with the pseudo-color images of restored HSI results further displayed in Figs. 6 and 10, the other methods more or less either suffer from some color distortions or cannot

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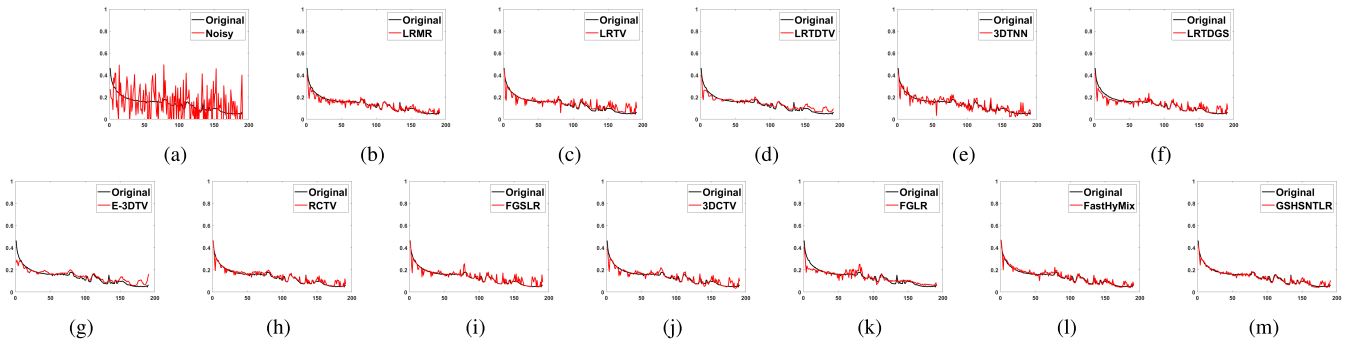


Fig. 9. Fitting results of spectral curves of different denoising methods on the simulated WDC dataset under Case 4 at spatial pixel (220, 93). Note that the black color denotes the spectral curve of the original HSI, and the red color denotes the spectral curve of restored HSI of different denoising methods, where the closer the red color fits the black color, the better the method preserves the spectral curve. (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

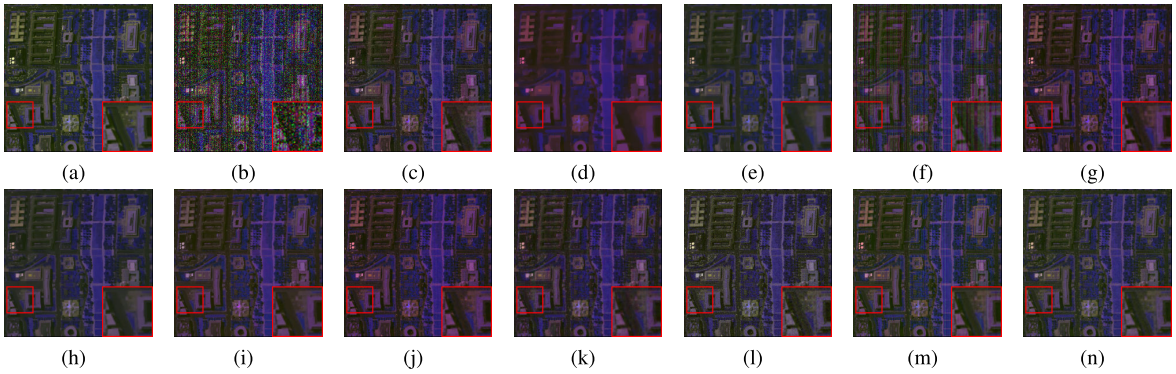


Fig. 10. Pseudo-color images of restored HSI results of different denoising methods on (32, 18, 100) bands of the simulated WDC dataset under Case 4. (a) Original image. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) 3DTNN. (g) LRTDGS. (h) E-3DTV. (i) RCTV. (j) FGSLR. (k) 3DCTV. (l) FGLR. (m) FastHyMix. (n) GSHSNTLR.

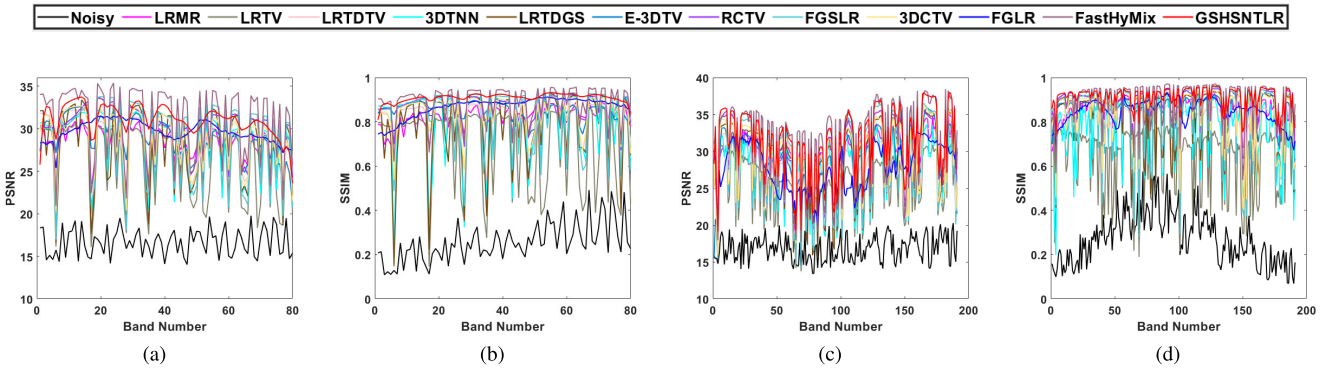


Fig. 11. PSNR and SSIM of each spectral band of different denoising methods on the simulated PaviaC and WDC datasets under Case 4. (a) PSNR of each spectral band on PaviaC. (b) SSIM of each spectral band on PaviaC. (c) PSNR of each spectral band on WDC. (d) SSIM of each spectral band on WDC.

preserve image structures and completely remove the mixed noise. Specifically, the proposed GSHSNTLR approach can better remove the mixed noise and preserve image structures with much less color distortions, whose color is much closer to that of the original HSI, showing much better spectral recovery performance.

What is more, Fig. 11 also displays the detailed PSNR and SSIM results of each spectral band of the restored HSI of different methods on the simulated PaviaC and WDC datasets under Case 4; the proposed GSHSNTLR approach can generally show higher, more stable, and robust PSNR and SSIM results than the other methods.

C. Denoising Experiments on Real HSI Datasets

More specifically, the subimage of GF-5 Shanghai dataset (size: $300 \times 300 \times 155$), the subimage of Indian Pines dataset (size: $145 \times 145 \times 206$), and the subimage of HYDICE Urban dataset (size: $207 \times 207 \times 210$) are adopted as the noisy HSI inputs for the denoising experiments on real HSI datasets, where the GF-5 Shanghai dataset is severely polluted by Gaussian noise, stripe noise, deadlines, impulse, and atmospheric noise, the Indian Pines dataset is severely polluted by dense stripe noise, deadlines, impulse noise, and Gaussian noise, and the HYDICE Urban dataset is seriously polluted by stripe

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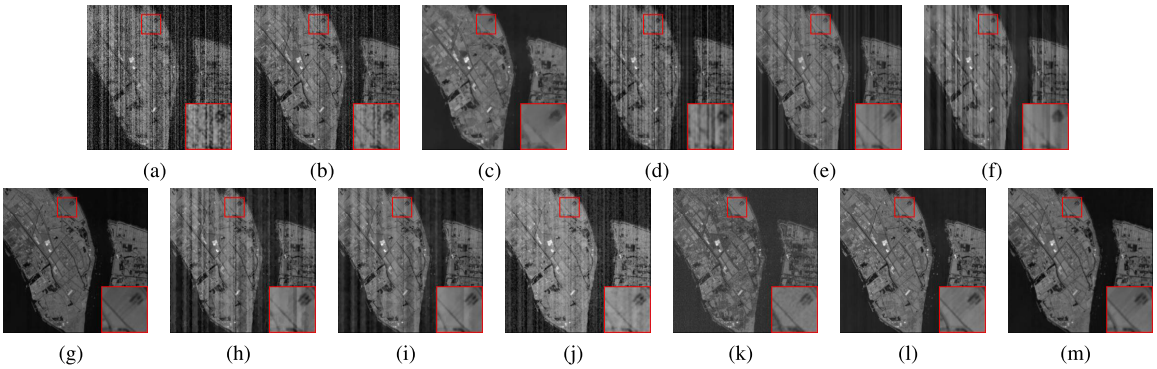


Fig. 12. Restored HSI results of different denoising methods on 155th band of the real GF-5 Shanghai dataset. (a) Real noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

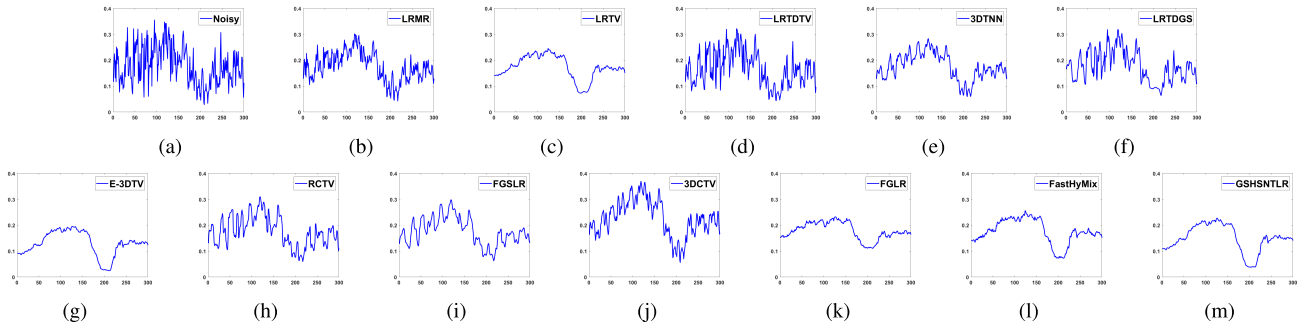


Fig. 13. Column mean curve results of different denoising methods on band 155 of the real GF-5 Shanghai dataset. (a) Real noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

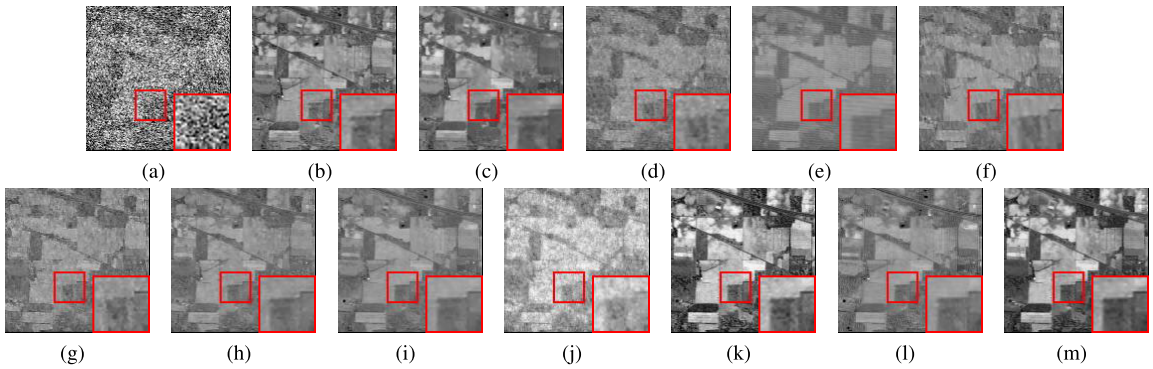


Fig. 14. Restored HSI results of different denoising methods on 106th band of the real Indian Pines dataset. (a) Real noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

noise, deadlines, water absorption noise, atmospheric noise, Gaussian noise, and some unknown noise.

Thus, to more comprehensively evaluate these denoising methods, Fig. 12 displays the visual denoising results of restored HSI on 155th band of the real GF-5 Shanghai dataset, and Fig. 13 further displays the column mean curve results of different denoising methods on band 155 of the real GF-5 Shanghai dataset. Besides, Fig. 14 displays the visual denoising results of restored HSI on the 106th band of the real Indian Pines dataset. To better illustrate and compare the spectral recovery performance of different methods, Figs. 15 and 16 further display the pseudo-color images of restored HSI results of different denoising methods of the real Indian Pines and HYDICE Urban datasets, respectively. Moreover, Fig. 17 displays the spectral curves of spatial pixel (17, 184) in the real HYDICE Urban dataset before and after denoising by different methods for further comparisons.

Clearly, by observing the restored HSIs and enlarged details shown in Figs. 12 and 14, LRMR, LRTV, LRTDTV, 3DTNN, LRTDGS, E-3DTV, RCTV, FGSLR, 3DCTV, FGLR, and FastHyMix either cannot completely reduce the stripe noise and other hybrid noise or cannot finely preserve image structures while smoothing some of the image structures. More clearly, the proposed GSHSNTLR approach can much better reduce the stripe noise and other hybrid noise simultaneously and more finely preserve image structures, which performs the best visual quality.

Furthermore, with the column mean curves of different denoising methods on band 155 of the real GF-5 Shanghai dataset displayed in Fig. 13, because the band 155 of the

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Fig. 15. Pseudo-color images of restored HSI results of different denoising methods on (201, 106, 20) bands of the real Indian Pines dataset. (a) Real noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

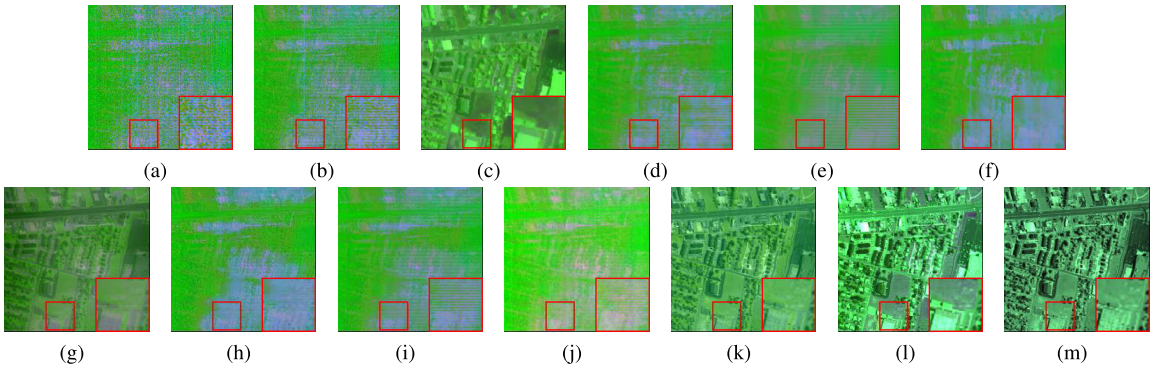


Fig. 16. Pseudo-color images of restored HSI results of different denoising methods on (108, 105, 150) bands of the real HYDICE Urban dataset. (a) Real noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

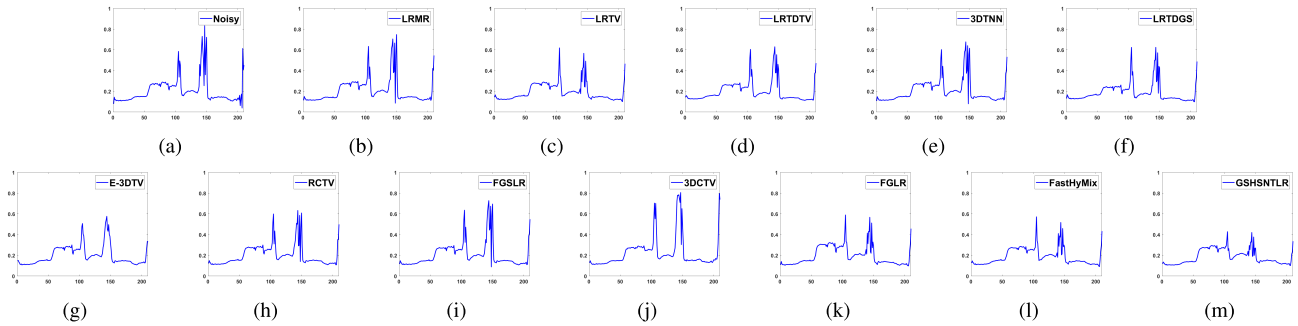


Fig. 17. Spectral curves of spatial pixel (17, 184) in the real HYDICE Urban dataset before and after denoising by different methods. (a) Real noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) 3DTNN. (f) LRTDGS. (g) E-3DTV. (h) RCTV. (i) FGSLR. (j) 3DCTV. (k) FGLR. (l) FastHyMix. (m) GSHSNTLR.

real GF-5 Shanghai dataset is severely polluted by stripe noise, Gaussian noise, deadlines, and other hybrid noise, the column mean curve of the noisy image has very large oscillations. As can be seen in Fig. 13, all the methods can smooth the column mean curves at different degrees, while still having some oscillations at different degrees. By much closer comparisons, the proposed GSHSNTLR approach shows the best smooth performance of the column mean curve.

Meanwhile, with the pseudo-color images of restored HSI results further displayed in Figs. 15 and 16, the other methods either suffer from some color distortions or cannot preserve image structures and completely remove the mixed noise. More specifically, the proposed GSHSNTLR approach can better remove the mixed noise and preserve image structures with much less color distortions, which shows much better spectral recovery performance.

Moreover, with the spectral curves of spatial pixel (17, 184) in the real HYDICE Urban dataset before and after denoising by different methods further displayed in Fig. 17, due to the existence of the hybrid noise, the spectral curve of real noisy image in Fig. 17(a) rapidly oscillates. Clearly, all these methods can suppress the oscillations at different degrees, where the proposed GSHSNTLR approach best suppresses the oscillations while better maintaining some minor changes in the spectral curve.

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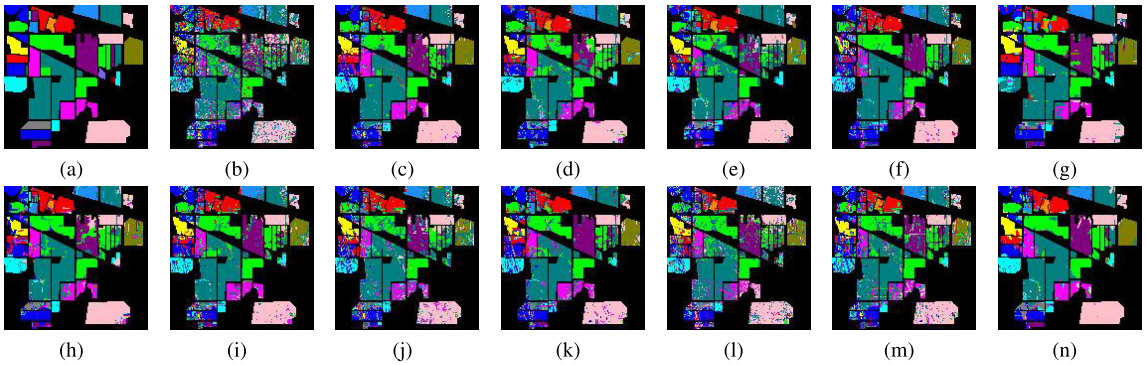


Fig. 18. Comparison of classification maps of different denoising methods on the real Indian Pines dataset. (a) Ground truth. (b) Noisy image. (c) LRMR. (d) LRTV. (e) LRTDTV. (f) 3DTNN. (g) LRTDGS. (h) E-3DTV. (i) RCTV. (j) FGSLR. (k) 3DCTV. (l) FGLR. (m) FastHyMix. (n) GSHSNTLR.

TABLE II
OA, AA, AND KAPPA VALUES OF DIFFERENT METHODS ON THE REAL INDIAN PINES DATASET

	Noisy	LRMR	LRTV	LRTDTV	3DTNN	LRTDGS	E-3DTV	RCTV	FGSLR	3DCTV	FGLR	FastHyMix	GSHSNTLR
OA	0.4583	0.8010	0.7601	0.6842	0.7174	0.8048	0.7968	0.7550	0.6648	0.7658	0.6569	0.6537	**0.8614**
AA	0.2777	0.5765	0.5922	0.5058	0.4552	0.5897	0.5954	0.5251	0.4610	0.5304	0.5136	0.4869	**0.6694**
kappa	0.3650	0.7719	0.7240	0.6318	0.6720	0.7750	0.7670	0.7181	0.6108	0.7305	0.6048	0.6019	**0.8412**

D. Comparisons of Classification on Real Indian Pines Dataset

More specifically, to analyze the restored HSI images of different denoising methods on the real Indian Pines dataset for the downstream applications, we particularly conduct a classification experiment and analyze their classification results by using the support vector machine (SVM) [57] algorithm to illustrate the classification effectiveness of these methods. As we know, the ground truth of the classification of Indian Pines dataset has 16 classes. In our experiments, we randomly choose the 10% of samples as training samples and the rest samples as testing samples for each class.

Then, the qualitative comparison of classification maps of different methods on the real Indian Pines dataset is displayed in Fig. 18. Moreover, Table II gives the quantitatively numerical evaluation results of overall accuracy (OA), average accuracy (AA), and kappa coefficient. Clearly, for the real Indian Pines dataset denoised by different denoising methods, the classification accuracy of the real noisy HSI can be further improved. As shown in Fig. 18, our proposed GSHSNTLR method gives the best classification performance visually. Meanwhile, Table II shows that our proposed GSHSNTLR method gives the best OA, AA, and kappa results. Thus, this classification experiment also illustrates the effectiveness of our proposed GSHSNTLR method.

E. Experimental Analysis

1) Analysis of Parameters: Furthermore, we discuss the influence of parameters of the proposed GSHSNTLR model (12) and (14), which includes five key parameters, i.e., the tradeoff regularization parameters β, α, and λ, and the penalty parameters η and μ.

Specifically, to comprehensively illustrate how these model parameters behave under different types of noise and across various datasets, we particularly investigate the joint influence

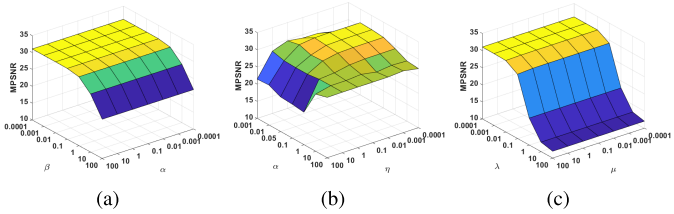


Fig. 19. Analysis of the parameters of the proposed GSHSNTLR method on simulated PaviaC dataset under Case 4. (a) MPSNR versus (β, α). (b) MPSNR versus (α, η). (c) MPSNR versus (λ, μ).

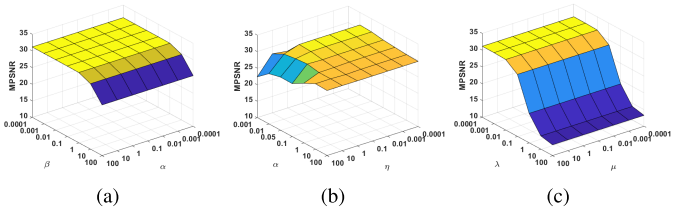
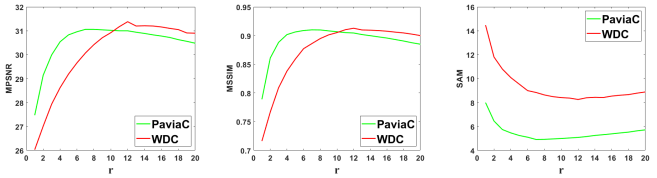


Fig. 20. Analysis of the parameters of the proposed GSHSNTLR method on simulated WDC dataset under Case 5. (a) MPSNR versus (β, α). (b) MPSNR versus (α, η). (c) MPSNR versus (λ, μ).

of these parameters β, α, η, λ, and μ of the proposed GSHSNTLR method on the simulated PaviaC dataset under Case 4 and the simulated WDC dataset under Case 5, and, respectively, display the results of MPSNR versus β, α, η, λ, and μ of GSHSNTLR in Figs. 19 and 20 for analysis.

As obviously displayed in Figs. 19 and 20, different settings of parameters β, α, η, λ, and μ can influence the denoising performance of the proposed GSHSNTLR method at different senses. Specifically, the proposed GSHSNTLR method reaches the optimal MPSNR results when we set $\beta \in [0.01, 1]$, $\alpha \in [0.01, 0.1]$, $\eta \in [0.01, 0.1]$, $\lambda \in [0.001, 0.01]$, and $\mu \in [0.0001, 0.001]$ in our experiments.

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(a) (b) (c)

Fig. 21. Analysis of the gradient subspace dimension r of the proposed GSHSNTLR method on the simulated PaviaC and WDC datasets under Case 4. (a) MPSNR versus r. (b) MSSIM versus r. (c) SAM versus r.

2) Analysis of the Gradient Subspace Dimension r: Furthermore, we analyze the influence of the gradient subspace dimension r of the proposed GSHSNTLR method, namely, the rank values of $\text{unfold}_3(\nabla_1\mathcal{X})$ (or $\nabla_1\mathbf{X}_{(3)}$), $\text{unfold}_3(\nabla_2\mathcal{X})$ (or $\nabla_2\mathbf{X}_{(3)}$), and $\text{unfold}_3(\nabla_3\mathcal{X})$ (or $\nabla_3\mathbf{X}_{(3)}$), where $\text{rank}(\nabla_1\mathbf{X}_{(3)}) = \text{rank}(\nabla_2\mathbf{X}_{(3)}) = \text{rank}(\nabla_3\mathbf{X}_{(3)}) = r$, which is clearly described in Remark 1 in Section II-B.

To this end, without loss of generality, we particularly investigate the effect of r of GSHSNTLR on the simulated PaviaC and WDC datasets under Case 4, and display the results of MPSNR, MSSIM, and SAM versus r of GSHSNTLR in Fig. 21 for analysis.

As obviously displayed in Fig. 21, different settings of r can influence the denoising performance of GSHSNTLR at different senses, and the proposed GSHSNTLR method reaches the optimal MPSNR, MSSIM, and SAM results when we set $r = 7$ for the simulated PaviaC dataset and $r = 12$ for the simulated WDC dataset in our experiments.

3) Analysis of Model Ablation: For the proposed GSHSNTLR model (12), it not only consists of the common prior term $\|\mathcal{S}\|_1$ of the sparse noise $\mathcal{S}$, but also consists of another two important nonconvex tensor low-rank prior terms, i.e., the LogTNN-based nonconvex tensor low-rank prior term of the gradient subspace-based RCT proposed in (7), and the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term $\|\mathcal{X}+\mathcal{B}\|_{\text{LogTNN}}$ proposed in (11). Thus, we further discuss how the above three prior terms behave in the proposed GSHSNTLR model (12). To this end, we consider the following variant models.

1) *Model 1:* GSHSNTLR without the prior term $\|\mathcal{S}\|_1$ of the sparse noise $\mathcal{S}$, i.e., GSHSNTLR with $\beta = 0$.
2) *Model 2:* GSHSNTLR without the LogTNN-based nonconvex tensor low-rank prior term of the gradient subspace-based RCT proposed in (7), i.e., GSHSNTLR with $\alpha = 0$.
3) *Model 3:* GSHSNTLR without the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term proposed in (11), i.e., GSHSNTLR with $\lambda = 0$.
4) *Model 4:* More specifically, to further validate the effectiveness of the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term $\|\mathcal{X}+\mathcal{B}\|_{\text{LogTNN}}$ proposed in (11), we particularly replace it in the proposed GSHSNTLR model (12) with the LogTNN-based nonconvex tensor low-rank prior term of only the stripe noise, i.e., $\|\mathcal{B}\|_{\text{LogTNN}}$, and obtain the following Model 4, that is:

$$\begin{aligned}(\hat{\mathcal{X}},\hat{\mathcal{S}},\hat{\mathcal{B}},\hat{\mathcal{U}}_i,\hat{\mathbf{V}}_i) = \underset{\mathcal{X},\mathcal{S},\mathcal{B},\mathcal{U}_i,\mathbf{V}_i}{\arg\min} & \frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2+\beta\|\mathcal{S}\|_1 \\ & + \alpha\sum_{i=1}^{3}\|\mathcal{U}_i\|_{\text{LogTNN}} + \lambda\|\mathcal{B}\|_{\text{LogTNN}} \\ \text{s.t.}\quad & \nabla_i\mathcal{X} = \mathcal{U}_i \times_3 \mathbf{V}_i \\ & \mathbf{V}_i^T\mathbf{V}_i = \mathbf{I}, \quad i = 1,2,3.\end{aligned} \tag{34}$$

5) *Model 5:* Moreover, based on the spectral low-rank property of HSI, to further validate the effectiveness of using the gradient domain for modeling the LogTNN-based nonconvex tensor low-rank prior term of the gradient subspace-based RCT proposed in (7) while still keeping the spectral low-rank constraints of HSI, we particularly choose the original domain instead of the gradient domain of HSI $\mathcal{X}$, and use the spectral low-rank decomposition of only HSI $\mathcal{X}$ for modeling, i.e., $\mathcal{X} = \mathcal{U} \times_3 \mathbf{V}$, $\mathbf{V}^T\mathbf{V} = \mathbf{I}$. Thus, based on the proposed GSHSNTLR model (12), we construct the following Model 5, i.e.,

$$\begin{aligned}(\hat{\mathcal{X}},\hat{\mathcal{S}},\hat{\mathcal{B}},\hat{\mathcal{U}},\hat{\mathbf{V}}) = \underset{\mathcal{X},\mathcal{S},\mathcal{B},\mathcal{U},\mathbf{V}}{\arg\min} & \frac{1}{2}\|\mathcal{Y}-\mathcal{X}-\mathcal{S}-\mathcal{B}\|_F^2+\beta\|\mathcal{S}\|_1 \\ & + \alpha\|\mathcal{U}\|_{\text{LogTNN}} + \lambda\|\mathcal{X}+\mathcal{B}\|_{\text{LogTNN}} \\ \text{s.t.}\quad & \mathcal{X} = \mathcal{U} \times_3 \mathbf{V} \\ & \mathbf{V}^T\mathbf{V} = \mathbf{I}.\end{aligned} \tag{35}$$

TABLE III
MODEL ABLATION ANALYSIS OF THE PROPOSED GSHSNTLR MODEL ON THE SIMULATED PAVIAC DATASET UNDER CASE 4

	MPSNR	MSSIM	SAM
Model 1: GSHSNTLR with $\beta = 0$	29.66	0.8841	6.1933
Model 2: GSHSNTLR with $\alpha = 0$	27.24	0.7866	9.7104
Model 3: GSHSNTLR with $\lambda = 0$	30.34	0.8880	6.5003
Model 4	30.50	0.8956	5.4018
Model 5	29.26	0.8856	6.5126
GSHSNTLR	**31.05**	**0.9100**	**4.9131**

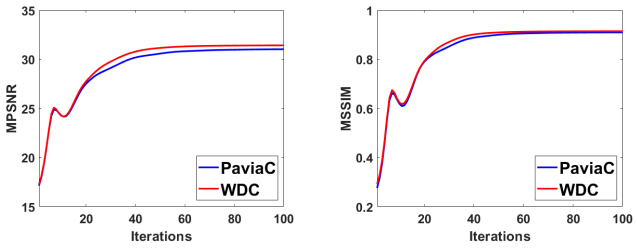
Without loss of generality, we investigate the denoising performance of the above five variant models as well as the proposed GSHSNTLR model by using the simulated PaviaC dataset under Case 4, and provide their MPSNR, MSSIM, and SAM results in Table III for analysis.

As obviously given in Table III, GSHSNTLR gives the best MPSNR, MSSIM, and SAM results. Thus, we can conclude that the prior term $\|\mathcal{S}\|_1$ of sparse noise, the LogTNN-based nonconvex tensor low-rank prior term of the gradient subspace-based RCT proposed in (7), and the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term proposed in (11) all influence the denoising performance at different senses, which all play important and indispensable roles in the proposed GSHSNTLR model (12) for HSI destriping and denoising. Moreover, by comparing the results of GSHSNTLR and Model 4, we can further demonstrate the superiority of the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank

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TABLE IV
COMPARISON OF RUNNING TIME (IN S) OF DIFFERENT DENOISING METHODS ON THE REAL DATASETS

Datasets	Size	LRMR	LRTV	LRTDTV	3DTNN	LRTDGS	E-3DTV	RCTV	FGSLR	3DCTV	FGLR	FastHyMix	GSHSNTLR
GF-5 Shanghai	$300 \times 300 \times 155$	54.2	365.5	198.2	244.9	141.8	65.6	18.4	544.6	519.2	14.1	36.6	202.5
Indian Pines	$145 \times 145 \times 206$	28.9	130.7	86.7	85.8	58.3	27.8	6.6	616.6	202.4	6.1	4.2	93.1
HYDICE Urban	$207 \times 207 \times 210$	48.8	218.4	145.2	183.0	103.2	47.5	9.6	2067.1	280.2	8.6	15.3	126.6



(a) (b)

Fig. 22. Analysis of convergence of GSHSNTLR on simulated HSI datasets of PaviaC and WDC under Case 4. (a) MPSNR versus iterations. (b) MSSIM versus iterations.

prior term proposed in (11). More specifically, by comparing the results of GSHSNTLR and Model 5, we can also further demonstrate the superiority of using the gradient domain for the proposed GSHSNTLR model.

4) Analysis of Convergence: Subsequently, we plan to analyze the convergence of the proposed GSHSNTLR algorithm in Algorithm 1. To this end, Fig. 22 representatively displays the results of MPSNR and MSSIM versus iterations of GSHSNTLR on the simulated HSI datasets of PaviaC and WDC under Case 4 for analysis. As evidently shown in Fig. 22, the values of MPSNR and MSSIM gradually rise with the increasing of iterations generally, subsequently become stable, and ultimately reach a convergent state, which assures the convergence of the proposed GSHSNTLR algorithm.

More specifically, we set $MaxIter = 100$ for the proposed GSHSNTLR algorithm in our experiments, which can enough guarantee the convergence of GSHSNTLR.

5) Analysis of Running Time: As given in Table IV, we, in particular, evaluate the performance of running time of different HSI denoising methods on the real GF-5 Shanghai, Indian Pines, and HYDICE Urban datasets. In Table IV, we clearly obtain that FastHyMix, FGLR, and RCTV are the fastest, the second-fastest, and the third-fastest methods, respectively, but they fail to effectively reduce the seriously hybrid noise. Moreover, the proposed GSHSNTLR method not only gives the optimal denoising performance of reducing the seriously hybrid noise, but also exhibits the acceptable performance of running time since the proposed GSHSNTLR model is relatively more complex.

6) Analysis of Computational Complexity: At last, we analyze the computational complexity of the proposed GSHSNTLR algorithm in Algorithm 1, which consists of the following parts.

1) The complexity of $\mathcal{S}^{(k+1)}$ via (16) at each iteration is $O(n_1n_2n_3)$.
2) The complexity of $\mathcal{B}^{(k+1)}$ via (19) at each iteration is $O(n_1n_2n_3)$.
3) The complexity of $\mathcal{A}^{(k+1)}$ via (21) at each iteration is $O(n_1n_2n_3(\log(n_1n_2n_3) + \min(n_1, n_2)))$.
4) The complexity of $\mathcal{U}_i^{(k+1)}$ via (25) at each iteration is $O(n_1n_2n_3(\log(n_1n_2n_3) + \min(n_1, n_2)))$.
5) The complexity of $\mathbf{V}_i^{(k+1)}$ via (28) at each iteration is $O(n_3r^2)$.
6) The complexity of $\mathcal{X}^{(k+1)}$ via (32) at each iteration is $O(n_1n_2n_3(\log(n_1n_2n_3)))$.
7) The complexity of $\mathcal{H}^{(k+1)}$ and $\mathcal{P}_i^{(k+1)}$ via (33) at each iteration is $O(n_1n_2n_3)$.

Consequently, the computational complexity of the proposed GSHSNTLR algorithm in Algorithm 1 at each iteration is totally $O(n_1n_2n_3(\log(n_1n_2n_3) + \min(n_1, n_2)) + n_1n_2n_3(\log(n_1n_2n_3)) + n_1n_2n_3 + n_3r^2)$.

V. CONCLUSION

This article proposes a novel and effective GSHSNTLR method for HSI destriping and denoising. Based on the mode-3 low-rank properties of the spatial and spectral gradients of HSI in the spectral mode, we apply the mode-3 low-rank decomposition of the gradients of HSI to obtain their RCTs and further explore the tensor low-tubal-rank properties of the RCTs in the gradient subspace. Thus, we propose the unified gradient subspace-regularized LogTNN-based nonconvex tensor low-rank prior term of the RCTs. Moreover, based on the structural speciality of stripe noise, which has strong tensor low-tubal-rank property, we particularly study the HSI-guided tensor low-rank modeling for the stripe noise, explore the tensor low-tubal-rank property of HSI plus stripe, and propose the unified HSI and stripe-coupled LogTNN-based nonconvex tensor low-rank prior term of HSI and stripe simultaneously. Subsequently, the ADMM method is used to optimize the proposed GSHSNTLR model efficiently. At last, extensive experimental results fully validate the denoising performance and superiority of GSHSNTLR.

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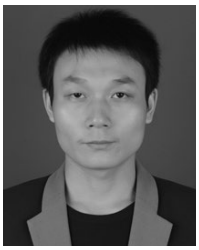
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